

Geometric Number Theory

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Algebra is the offer made by the devil to the mathematician. The devil says: "I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine."

— Sir Michael Atiyah

1 Introduction

This set of notes is meant to be an exposition of a proof of one of the most fundamental theorems in algebraic number theory: unique prime factorization of ideals in a Dedekind domain. It is a generalization of unique prime factorization over $\mathbb{Z}$ which works over a general class of rings which satisfy some specific axioms (Dedekind domains). This theorem is more of a theorem-definition as Dedekind engineered the axioms of his namesake domain precisely to obtain unique prime factorization of ideals; thus understanding the theorem should come with a good understanding of the definition. However, the "standard" proof, in my opinion, does not elucidate why these axioms are the right ones. These notes will present an alternative proof which approaches the theory from a more geometric standpoint where the three axioms feel more natural.

A good reference for this material is [2] where I first learned the proof. This document is essentially a condensed form of the first section of the book with some additional intuitions I learned elsewhere sprinkled in. The proofs of some of the algebraic facts are omitted as the focus is on the geometric aspects, but they can all be found in [2] which contains a full proof of the theorem. Facts used from algebraic geometry can be found in [1], [5], or just about any algebraic geometry textbook that discusses schemes.

2 Overview

2.1 Why Ideals

Algebraic number theory started by trying to generalize the notion of factorization to other "number systems" such as $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{2}]$, or $\mathbb{Z}[\sqrt{-5}]$. The main theorem is stated pretty easily: in a "number system" (some suitable ring which acts like the integers), every ideal can be represented as the product of prime ideals, unique up to permutation. Some simple examples to get started: in $\mathbb{Z}[i]$, 2 is not prime, but $2 = (1+i)(1-i)$, which can be "idealized" by parenthesizing 2 and treating everything as an ideal. In $\mathbb{Z}[\sqrt{-5}]$, $(2) = (2, 1 + \sqrt{-5})(2, 1 - \sqrt{-5})$.

In my opinion, this statement raises more questions than it answers: sure the statement "feels like" unique prime factorization over the integers, but why work with ideals, and what is the right notion of a number system (in fact, the definition of a Dedekind domain is basically cooked up to make unique prime factorization work). I think that in learning about ideals, it is best to trace how they came about historically. It is in fact intimately tied to the question of unique prime factorization.

Let's look more closely at an example where the uniqueness of prime factorization fails. In $\mathbb{Z}[\sqrt{-5}]$, $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. Now, in $\mathbb{Z}$, $12 = 2 \cdot 6 = 3 \cdot 4$, but this is not a problem because 6 and 4 can be decomposed further yielding $12 = 2^2 \cdot 3$. However, the case here is more extreme since none of 2, 3, $1 + \sqrt{-5}$, or $1 - \sqrt{-5}$ can be broken down further. The argument is fairly direct: consider the norm function $N(\alpha) = |\alpha|^2$ i.e. the square of the complex absolute value of some $\alpha \in \mathbb{Z}[\sqrt{-5}]$. Why? We can transfer our knowledge of decomposition in $\mathbb{Z}$ to $\mathbb{Z}[\sqrt{-5}]$ as follows: it isn't too hard to show $N(\alpha\beta) = N(\alpha)N(\beta)$ and that $N(\alpha)$ is always an integer (this is why we take the square of the absolute value rather than the absolute value). We can now use unique factorization in the integers to say something substantial. Say $2 = xy$ for $x, y \in \mathbb{Z}[\sqrt{-5}]$. Then, $N(x)N(y) = N(xy) = N(2) = 4$ so that $N(x)$ is 1, 2, or 4 and the same for y . Now, if $x = a + b\sqrt{-5}$, $N(x) = a^2 + 5b^2$. We see then that it is impossible for the norm to be 2, while norm 1 is possible iff $x = \pm 1$ and 4 is only possible if $x = \pm 2$, but these don't break down 2 at all. It can be similarly checked that 3, $1 + \sqrt{-5}$, and $1 - \sqrt{-5}$ are all irreducible as well.

Enter Kummer. We have shown that in the ring $\mathbb{Z}[\sqrt{-5}]$, there are truly two different distinct factorizations of 6 into irreducible factors. Kummer thought, what if we added numbers to the ring which would forcibly break the factors down further, for example, add a number that is a factor of 2 and $1 + \sqrt{-5}$ which could allow us to break the factorization further like we did for 12. He called these "ideal numbers," and in fact, this is where we get the name ideal from! As nice of a solution this would be, adding new numbers may fix some factorizations like for 6, but could cause problems with messing up prime factorization elsewhere. Dedekind came up with a brilliant solution: instead of adding an ideal number, try to capture "the essence of what the number would be". Say we want to add a number which is simultaneously a factor of 2 and $1 + \sqrt{-5}$. There may be multiple candidates, after all, we're free to choose basically any complex number. Ideally, we want to add the smallest possible change to our ring, so we want a number which is a factor of 2 and $1 + \sqrt{-5}$, but it shouldn't be a factor of another number, say 5, since we want to use it to factorize 6. Instead, we can look at the minimal set of numbers that it can be factors of. Say α is our ideal number and S is the set of numbers in $\mathbb{Z}[\sqrt{-5}]$ which are multiples of α . By basic properties of multiplication, if $\alpha|a$ and $\alpha|b$, then $\alpha|(a+b)$ and also for any $r \in \mathbb{Z}[\sqrt{-5}]$, $\alpha|(a \cdot r)$. Hence, S is closed under addition and scaling by arbitrary elements of our ring. But we recognize this as the conditions for an ideal! Hence, ideals can be thought of as shadows of ideal numbers that remain in the ring we care about. Dedekind's brilliance was to show that this was enough to recover an as-good-as-can-be-hoped analogue to prime factorization over the integers. In finding the generalization of prime factorization, he was simultaneously led to define Dedekind domains as the right generalization of the integers, which is the type of ring over which unique prime factorization of ideals works. I think this example serves to remind us that mathematics isn't a clean cut process and really is a matter of trial and error.

2.2 A Sketch of the Proof

Unique prime factorization over $\mathbb{Z}$ is usually proven (at a very high level) as follows: Euclid's algorithm gives a way to find the greatest common factor of two numbers by adding and scaling, i.e., the gcd of two numbers lies in the ideal generated by them. From this, it follows that every ideal in $\mathbb{Z}$ is principal i.e. $\mathbb{Z}$ is a PID. Using some algebraic machinery, it can be shown that being a PID implies having unique factorization into primes.

The key step of this proof is Euclid's algorithm. This is where this proof fails in general - Euclid's algorithm works by making smaller and smaller numbers that divide both initial numbers until the process can go no further, but for general "number rings" like $\mathbb{Z}[\sqrt{-5}]$, there is no natural notion of "size". Indeed, if there were, $\mathbb{Z}[\sqrt{-5}]$ would be a PID, but we showed it wasn't.

There is however one more context in which we have unique factorization. This is the unique factorization of polynomials over an algebraically closed field, $k[x]$. This has a very nice structure: every polynomial $f(x)$ has a root in an algebraically closed field (essentially by definition), so we take such a root and extract it from f , and then we end up with a polynomial of smaller degree. Induct until we get all the roots. This has a nice geometrical interpretation: $f(x)$ can be interpreted as a function

from k to k (both interpreted as "lines"), and k being algebraically closed says that there is at least one point a where this function vanishes. We can divide not just by $x - a$, but in fact by $(x - a)^n$ where n is the multiplicity of the zero at a . The function now does not vanish at a and we keep dividing until the function no longer vanishes. This implies the function is constant, and thus we have completely factorized our polynomial.

The general proof of unique prime factorization will actually be to change our mindset to view number rings like $\mathbb{Z}$ and $\mathbb{Z}[\sqrt{5}]$ as being sets of not "numbers", but "functions" on some curve. The points of the curve will turn out to be the prime ideals we are looking for. For some number, we will find points at which this number, or rather "function" vanishes, which corresponds to prime ideals dividing our number. We can "divide out" at this prime and get a number of "smaller degree", and eventually end up at 1. Now would be a good time to finally introduce

2.2.1 The Axioms of a Dedekind Domain

As it is usually stated, the definition of a Dedekind domain is as follows:

Definition 1 (Dedekind Domain). *An integral domain A is a Dedekind Domain if it in addition satisfies the following axioms:*

- A is Noetherian
- Every nonzero prime ideal of A is maximal
- A is integrally closed

The last one requires its own

Definition 2 (Integrally Closed). *A subring A of B is said to be integrally closed in B if the following is true: if there is some $f(x) \in A[x]$ such that there is some $b \in B$ for which $f(b) = 0$, then in fact $b \in A$. A ring A is said to be integrally closed if it is integrally closed in its field of fractions.*

The integral closure of a ring A in B is the set of all elements of B which are roots to some polynomial equation with coefficients in A . The integral closure of A is its integral closure in its field of fractions.

Concretely, if A is integrally closed, it means that if $f(x) \in A[x]$ is some polynomial that has a root in $\text{Frac}(A)$, that root in fact lies in A . A counterexample to this is $A = \mathbb{Z}[\sqrt{-3}]$ where $\zeta_3 = \frac{-1+\sqrt{-3}}{2}$ is a root of $x^2 + x + 1 \in A[x]$, but is not itself in A . In fact, as we will see later, $\mathbb{Z}[\zeta_3]$ is the integral closure of A .

Now for some geometric interpretation of the axioms. These are only previews to showcase the big picture and will likely not make much sense now. The goal of the notes is to explain each of them in detail. Thus, I recommend to keep coming back to this part when things start to sound familiar down below. First, the Noetherian condition is a reasonable finiteness condition. The condition of every nonzero prime ideal being maximal can be restated as saying that A has Zariski dimension 1. This notion will be explained later, but it says that the elements of A can be interpreted as functions on some one dimensional curve $\text{Spec}(A)$, to be constructed later. This is similar to how $k[x]$ (which is a Dedekind domain) is a collection of functions on the one dimensional line k (which "is" $\text{Spec}(k[x])$), c.f. Hilbert's Nullstellensatz).

The condition that A is an integral domain says two things as failure to be an integral domain can happen because of nilpotent elements, or zero divisors. That there are no zero divisors means that this curve has one "component" (as opposed to the solutions of $xy = 0$ in $\mathbb{R}^2$ being the "union of two lines"). That this curve has no nilpotent elements means that this curve is "reduced", which is a more subtle notion and will be explained later.

These seem somewhat reasonable, but definitely the strangest condition is that of being integrally closed. It turns out that this has the geometric interpretation of the curve being nonsingular! This is precisely in the sense that $y^2 = x^3 - x^2$ has a singularity at the origin (graph it). The reason it is so strange is because being nonsingular is really a local condition that must be checked at every

point, and it turns out that being integrally closed is a global condition which locally turns out to be equivalent to nonsingularity.

Now for a sketch of the proof: an element a of a Dedekind domain A should be considered as a function on the curve $\text{Spec}(A)$. Looking locally at each point (prime ideal), we can find how much this function vanishes (in the sense of some multiplicity) at each point of $\text{Spec}(A)$. Putting together these multiplicities, we can write $(a) = \mathfrak{p}_1^{m_1} \dots \mathfrak{p}_n^{m_n}$ where the $\mathfrak{p}_i$ are where a vanishes to multiplicity m_i . This is analogous to how we can almost write $f(x) = (x - a_1)^{m_1} \dots (x - a_n)^{m_n}$, but only up to a constant multiple, which anyways yields $(f(x)) = (x - a_1)^{m_1} \dots (x - a_n)^{m_n}$ as an equation of ideals. It turns out this process does not care whether the initial ideal was principal, and thus will work for all ideals.

Where do the axioms come in? The fact that $\text{Spec}(A)$ is 1 dimensional means that a function vanishes at a discrete set of points, as opposed to $x^2 + y^2 - 1 = 0$ cutting out a circle in $\mathbb{R}^2$. Being Noetherian means that there are finitely many prime factors - if we keep dividing out by primes, we eventually end up at the unit ideal, or alternatively, "a function cannot vanish at infinitely many points".

The nonsingularity is a bit more subtle; it guarantees that the degree of vanishing is well defined. For example, consider the curve $xy = 0$ in $\mathbb{R}^2$, the union of the axes. It should seem reasonable that $(x - 1)^n$ vanishes to degree n at $(1, 0)$. In fact, any function which vanishes at $(1, 0)$ can be written as $(x - 1)^m g(x, y)$ where $g(1, 0) \neq 0$. But the behavior of functions at $(0, 0)$ is more subtle. If we have two functions that vanish to the same order at a point, their ratio should evaluate to a nonzero number at that point. Suppose then we say that x and y both vanish to order 1 at $(0, 0)$. Then, $\frac{x}{y}$ should evaluate to a nonzero number at the origin, but is instead undefined! Alternatively said, there is no (polynomial) $g(x, y)$ such that $y = xg(x, y)$. Thus x and y "do not vanish the same way". The subtlety is that $xy = 0$ is singular at the origin. Nilpotency also causes similar issues in defining multiplicity.

Axioms are great, but counterexamples to them are more helpful for understanding. Here is a list of counterexamples for each of the axioms of a Dedekind domain:

- Integral Domain (zero divisors): $\mathbb{Z} \times \mathbb{Z}$. Geometrically, this is a union of two copies of $\text{Spec}(\mathbb{Z})$, and thus reducible.
- Integral Domain (nilpotency): $\mathbb{Z}[x]/(x^2) = \{a + bx | x^2 = 0\}$. Geometrically, this is an "infinitesimal deformation" of $\text{Spec}(\mathbb{Z})$.
- Noetherian: $\bar{\mathbb{Z}}$, the absolute ring of integers, i.e., all algebraic numbers (zeros of monic equations in $\mathbb{Z}$) in $\bar{\mathbb{Q}}$, alternatively said, the integral closure of $\mathbb{Z}$ in $\bar{\mathbb{Q}}$. The going up and down theorems show that this is still a dimension 1 ring, it is clearly integral as it is a subring of a field, and is integrally closed by definition. The preimage of a prime $(p) \in \text{Spec}(\mathbb{Z})$ under the projection $\text{Spec}(\bar{\mathbb{Z}}) \rightarrow \text{Spec}(\mathbb{Z})$ is infinite, and thus p "vanishes at infinitely many points".
- Zariski Dimension 1: $\mathbb{Z}[x]$. This has Zariski dimension 2, and thus $\text{Spec}(\mathbb{Z}[x])$ is "like a plane". The zero locus of the ideal (x) is a copy of $\text{Spec}(\mathbb{Z})$, which is a whole line, and thus x vanishes at infinitely many points.
- Integrally Closed: As was stated earlier, $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed. This example will be analyzed much later, but the singularity is at the prime (2) .

3 Language of Schemes

The purpose of this section will be to introduce the language of schemes to talk about the integers. The main takeaway is that the integers should not be thought of as "points," but instead as a set of functions on a curve. The underlying ideas are that there is a very strong duality between algebra (i.e. function theory) and geometry. The subject so aptly named algebraic geometry essentially asks the question "how much geometry can be extracted from algebra?" On one hand, geometric questions which may seem initially intractable are transformed into algebraic ones where they can be attacked

systematically. Conversely, and what will be useful for us, one can utilize geometric intuition to gain deep insights into algebraic objects.

3.1 A Motivating Example

This example will showcase a relatively simple but surprising way that the geometry of a space can be extracted out of the functions on that space.

Consider the closed unit interval $X = [0, 1]$ and smooth $\mathbb{R}$ -valued functions on the interval, $R = C^\infty(X)$. Since R is closed under (pointwise) addition and multiplication and has a unit, it forms a ring. We can try and study R algebraically and try and classify its ideals. For reasons that likely seem mysterious, we would actually like to classify maximal ideals. First, we claim that for any closed subset $Y \subset X$, the set $I(Y) = \{f | f(x) = 0, \forall x \in Y\} \subset R$ (all functions which vanish on Y) is an ideal of R . Indeed, it is easy to verify that it is closed under addition and scaling by arbitrary elements of R .

Actually, it can be shown that for any proper ideal $J \subset R$, there is some $Y \subset X$ such that $J \subset I(Y)$, or phrased more geometrically, there is some subset Y such that every function in J vanishes on Y . Suppose on the contrary that there are functions f_i with supports covering X . Alternatively said, define $V(f) = \{x \in X | f(x) = 0\}$ (the vanishing locus of f), then we have that $\bigcap V(f_i) = \emptyset$. Then, since X is compact and $X \setminus V(f_i)$ is open for each i , we can choose finitely many f_i whose supports cover X , say $f_1, \dots, f_n$. Then, it can be checked that $F(x) = \sum_{i=1}^n f_i^2(x)$ is everywhere strictly positive on X . Indeed, at every $x \in X$, at least one f_i is nonzero and taking the sum of squares ensures it is strictly positive. In particular, $F(x)$ is never 0 so $\frac{1}{F(x)} \in R$. As J is an ideal and $F \in J$, $F \times \frac{1}{F} = 1 \in J$ which implies $J = R$.

It can be easily verified that if $Y \subset Y'$, $I(Y) \supset I(Y')$, in other words I is an order reversing map from subsets to ideals. This is surprisingly difficult to remember at first, so the way I like to think of it is any function that vanishes on Y' also vanishes on any smaller subset. From here, it logically follows that the maximal ideals of R are of the form $I(x) := I(\{x\})$ for $x \in X$. Indeed, any ideal J is contained in some $I(Y)$ and for any $x \in Y$, $I(Y) \subset I(x)$ (any function vanishing on Y vanishes on x). If $I(x)$ were contained in some larger ideal I' , then by definition of $I(x)$, I' would contain some function not vanishing on x and by a similar argument to the above, $I' = R$.

Thus, we have come to the remarkable conclusion that the *points* $x \in X$ are in 1-1 correspondence to the *maximal ideals* $I \subset R = C^\infty(X)$! Now would be a good time to mention that $R/I(x_0) \cong \mathbb{R}$ for $x_0 \in X$. In fact, the map sends $f(x) \in R$ to $f(x_0)$ since $f(x) - f(x_0)$ is a function which vanishes at x_0 (the so-called remainder theorem). Thus, not only is the quotient a field, but is naturally identified with evaluation of the function at the point that it corresponds to.

Now, in theory, if I handed you R as a ring without telling you that it was the ring of functions on X , you could recover the points of X . This may seem like a very strange thing to do since R is quite an unwieldy ring. You may also note that we never actually used any properties of functions of R too much stronger than continuity in a serious manner. Thus maybe we can associate different possible "rings of functions" (e.g. continuous or differentiable functions) which can also recover the points of X . You may also object that this is not a very interesting result since we can do exactly the same argument on any compact manifold, and every compact manifold has uncountably many points, so the result is that $R = C^\infty(X)$ just has uncountably many maximal ideals for whatever X we choose. What would be more interesting would be if we could recover some of the *topology* of X from R .

In fact, if we change perspectives from real manifolds to complex manifolds, and consider the much more restricted class of holomorphic functions, all these complaints would be resolved. It turns out in fact that holomorphic functions on compact complex manifolds are actually quite "small", usually just some variant of quotients of polynomial rings over $\mathbb{C}$ i.e. something which can actually be studied algebraically. The statement that maximal ideals correspond to points still holds, but now is a tractable problem to solve algebraically. Prime ideals are also important here since they correspond to complex submanifolds (actually, "subvarieties" which are zero sets of a collection of functions). The prime ideals of the ring of real valued functions however are not so well-behaved.

In fact much of the topology of a complex manifold can be recovered from the ring of functions. For example, a “complex curve” such as the (projectivization of the) solution set to $y^2 = x^3 - x$ in $\mathbb{C}^2$ is a 1-dimensional complex manifold and thus an orientable 2-dimensional real manifold, and hence has a genus. The genus of the curve can actually be computed purely algebraically. More generally, the entire cohomology of such a manifold can be computed purely algebraically by analyzing the functions closely enough. These results fall into the theory of algebraic geometry proper and are beyond the scope of this paper, but hopefully give some insight as to how functions can be useful in studying geometry. In case you were hoping for a similar correspondence to hold for real manifolds, there is a notion of Gelfand duality which puts more structure on the space of real valued functions to put more control on them and sets up a similar theory to the complex case.

3.2 Spectra of Rings

Now we have seen that in some favorable cases, if we are given a ring of functions on a topological space, we can recover much of the information of that space by studying the ring alone. If someone handed you such a ring, you could recover the points and maybe even the geometry of the space it came from. Mathematicians, seeking unity of formalism in everything, thought to do this for arbitrary rings. That is, given *any* ring, even $\mathbb{Z}$, can we think of it as the space of functions on some geometric object? The answer is “Yes! Just construct the geometric object from the ring.”

Motivated from the above discussion, given a ring R let’s define the associated geometric object, a set called the maximal spectrum of R :

$$\text{Specm}(R) = \{\mathfrak{m} \mid \mathfrak{m} \text{ is a maximal ideal of } R\}.$$

For $\mathbb{Z}$, there would be one point (p) for each prime p of $\mathbb{Z}$. We should have that $\mathbb{Z}$ be considered functions on the space $\text{Specm}(\mathbb{Z})$, but in what sense is this true? Usually, we think of functions as taking a point in some geometric space as a domain and having a value in some algebraic object like the field $\mathbb{R}$ or $\mathbb{C}$. Recall from the previous section that the quotient map $C^\infty(X)/I(x_0) \cong \mathbb{R}$ is really just the evaluation map $f \mapsto f(x_0)$. This suggests that we can define evaluations of these “functions” as being the images under quotient maps. That is, for an element r in a ring R and a maximal ideal $(\mathfrak{m}) \subset R$, we should consider the “value” of r at $\mathfrak{m}$ as being its image in the field $R/\mathfrak{m}$. For example, say we have the element $12 \in \mathbb{Z}$. This should be thought of as a function which evaluates to $0 \in \mathbb{Z}/2\mathbb{Z}$ at (2) , to $2 \in \mathbb{Z}/5\mathbb{Z}$ at (5) , and to $5 \in \mathbb{Z}/7\mathbb{Z}$ at (7) . Note that the codomain here changes from point to point, but at least is always a field.

The spirit of modern mathematics is to think categorically, and so it is not enough just to specify the geometric objects, but we should also define morphisms between these geometric objects. But what should these look like? To get an idea, let’s first return to manifold theory and look at how morphisms between manifolds affect their function spaces. For manifolds X and Y , a map $f : X \rightarrow Y$ induces the pullback map $f^* : C^\infty(Y) \rightarrow C^\infty(X)$ as $f^*(g) = g \circ f$. This is essentially the same as the dual of a vector space. A nice visual for this is to interpret g itself as a map from Y to $\mathbb{R}$ and then f^* acts as follows:

$$\begin{array}{ccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & \mathbb{R} \\ & \searrow & \swarrow & \nearrow & \\ & & f^*g = g \circ f & & \end{array}$$

Now our goal is to recover geometry from algebra. That is, given a morphism $F : C^\infty(Y) \rightarrow C^\infty(X)$, can we construct some $f : X \rightarrow Y$ such that $F = f^*$? If we can do this, we can transfer this construction to Specm since we know what morphisms of rings look like. We need to construct a way to take some $x \in X$ and associate a $y \in Y$ given the data of F . We already have a hint since we have a correspondence between points of our space and maximal ideals of our rings. Thus, say x corresponds to $\mathfrak{m}_x \subset C^\infty(X)$, which recall is the functions vanishing at x , and likewise y corresponds to $\mathfrak{m}_y \subset C^\infty(Y)$. The key insight is that for $f : X \rightarrow Y$, if g vanishes at y , $f^*(g)$ vanishes at x (think about the diagram above). Thus $f^*(\mathfrak{m}_y) \subset \mathfrak{m}_x$. If multiple points x_1, x_2, x_3 , say, all map to y , we would have that $f^*(\mathfrak{m}_y) \subset \mathfrak{m}_{x_1} \mathfrak{m}_{x_2} \mathfrak{m}_{x_3}$ since any function vanishing at y will pullback to a function

vanishing at x_1, x_2, x_3 (a priori, it is the intersection, but use Chinese remainder theorem). Since f^* should correspond to F , we should have that $F(m_y) = m_x$ i.e. $m_y \subset F^{-1}(m_x)$, but as m_y is maximal, this must mean that $m_y = F^{-1}(m_x)$.

We can now describe how to construct an appropriate $f : X \rightarrow Y$ given a morphism of rings $F : C^\infty(Y) \rightarrow C^\infty(X)$. Given some $x \in X$, define $f(x)$ as follows: x corresponds to some $\mathfrak{m}_x$ in $C^\infty(X)$, so consider the ideal $F^{-1}(\mathfrak{m}_x) \subset C^\infty(Y)$. If this is a maximal ideal, say $\mathfrak{m}_y$, define $f(x) = y$. Note that we need the algebraic map F to satisfy the property that the preimage of any maximal ideal is maximal. Moreover it is not clear that the defined map f is even continuous, let alone differentiable or smooth. Again, both of these problems can be proven away if we consider complex manifolds instead. The point of the preimage of a maximal ideal being maximal will rise up again in a moment.

Motivated by the above, we would like to define morphisms between maximal spectra as follows: a ring homomorphism $\varphi : S \rightarrow R$ gives rise to a morphism $f : \text{Specm}(R) \rightarrow \text{Specm}(S)$ by, $\forall \mathfrak{m} \in \text{Specm}(R)$, $f(\mathfrak{m}) = \varphi^{-1}(\mathfrak{m}) \in \text{Specm}(S)$. We define a morphism between maximal spectra to be a map which can be obtained in such a way. The way to think of this is that even though we define the map using a ring homomorphism, it is really a map of spaces which induces that ring homomorphism as the pullback on functions. This definition does not quite work, but let's look at some examples.

Example 1: $\text{Specm}(\mathbb{Z}/p\mathbb{Z})$ for a prime p is a single point as $\mathbb{Z}/p\mathbb{Z}$ is a field and has a unique maximal ideal. The quotient map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$ induces a map between spectra $F : \text{Specm}(\mathbb{Z}/p\mathbb{Z}) \rightarrow \text{Specm}(\mathbb{Z})$ which takes the single point $(0) \in \text{Specm}(\mathbb{Z}/p\mathbb{Z}) \mapsto \varphi^{-1}((0)) = (p) \in \text{Specm}(\mathbb{Z})$.

Example 2: It can be easily checked that the maximal ideals of $\mathbb{Z}/m$ are the ideals $(p) \subset \mathbb{Z}/m$ for a prime number $p \mid m$. The quotient map $\mathbb{Z} \rightarrow \mathbb{Z}/m$ induces a map between spectra $F : \text{Specm}(\mathbb{Z}/m) \rightarrow \text{Specm}(\mathbb{Z})$. This map should be seen as the inclusion of the discrete, finite set of points that make up $\text{Specm}(\mathbb{Z}/m)$ into the set of discrete points of $\text{Specm}(\mathbb{Z})$.

Example 3: $\text{Specm}(\mathbb{Z}[i])$ consists of 3 types of ideals: $(1+i)$, (p) for $p \cong 3 \pmod{4}$ and $(a+bi)$ where $a^2 + b^2 = p$ for $p \cong 1 \pmod{4}$ (this will be shown later). The inclusion $\mathbb{Z} \rightarrow \mathbb{Z}[i]$ induces a map $\text{Specm}(\mathbb{Z}[i]) \rightarrow \text{Specm}(\mathbb{Z})$ which takes an ideal $\mathfrak{p} \subset \mathbb{Z}[i]$ to $\mathfrak{p} \cap \mathbb{Z}$.

Example : Let $X = \text{Specm}(\mathbb{C}[x, y])$. Hilbert's Nullstellensatz gives us an identification of pairs (points, classically) $(z, w) \in \mathbb{C}^2$ and points (maximal ideals) $(x-z, y-w) \in \text{Specm}(\mathbb{C}[x, y])$. As $\mathbb{C}$ is a field, $\text{Specm}(\mathbb{C})$ is a single point. For a morphism $\varphi : \mathbb{C}[x, y] \rightarrow \mathbb{C}$, the kernel is a maximal ideal, and thus a point $(x-z, y-w)$. This induces a map $F : \text{Specm}(\mathbb{C}) \rightarrow \text{Specm}(\mathbb{C}[x, y])$ so that $F((0)) = \varphi^{-1}(0) = (x-z, y-w)$. There is one such map for each $(z, w) \in \mathbb{C}^2$, so unlike in the case for $\mathbb{Z}$, the morphism is not uniquely determined by the image ring since there are many different possible morphisms. Classically (under the Nullstellensatz correspondence), this corresponds to mapping a point to the point $(z, w) \in \mathbb{C}^2$. This is a very simple thing to do geometrically, but is more nuanced in the algebro-geometric setting since it comes from a morphism of rings.

Example : As we have seen, maps from rings R to fields induce maps from points to $\text{Specm}(R)$. We have a map $\varphi : \mathbb{Z} \rightarrow \mathbb{Q}$. Again, $\text{Specm}(\mathbb{Q})$ is a single point (0) , but $\varphi^{-1}(0)$ is $(0) \subset \mathbb{Z}$, which is not a maximal ideal, and thus not a point in $\text{Specm}(\mathbb{Z})$. A similar phenomenon happens looking at the inclusion $\mathbb{C}[x] \rightarrow \mathbb{C}(x)$.

The issue is that for a morphism $R \rightarrow S$ to define a map $\text{Specm}(S) \rightarrow \text{Specm}(R)$, we need the preimage of every maximal ideal to be maximal. This is not always the case, as in the inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$. However, we can always say that the preimage of a maximal ideal is at least a prime ideal, or better, the preimage of a prime ideal is prime: given $\varphi : R \rightarrow S$ if $\mathfrak{p}$ is a prime ideal of S , we get an induced map $R \rightarrow S/\mathfrak{p}$ which has kernel $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$, so it factors through the injection $R/\mathfrak{q} \rightarrow S/\mathfrak{p}$. As $S/\mathfrak{p}$ is integral ($\mathfrak{p}$ is prime), $R/\mathfrak{q}$, being a subring of $S/\mathfrak{p}$, is also integral, so $\mathfrak{q}$ is prime. Thus, being prime is better than being maximal in the sense that the pullback of a prime ideal is always prime. In fact, it is the smallest set of ideals which contain maximal ideals and satisfy this property for all morphisms. This motivates the definition:

Definition 3 (Affine Scheme). Given a ring R , the affine scheme $\text{Spec}(R)$ (more precisely its underlying set) is the collection of prime ideals $\mathfrak{p} \subset R$. Given two affine schemes $\text{Spec}(R)$ and $\text{Spec}(S)$, a map $F : \text{Spec}(R) \rightarrow \text{Spec}(S)$ is a map induced by a map of rings $\varphi : S \rightarrow R$ via $F(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p}) \in \text{Spec}(R)$ for $\mathfrak{p} \in \text{Spec}(S)$.

Example 1: If K is a field, the only prime ideal of K is (0) , so $\text{Spec}(K)$ consists of a single point. In particular, $\text{Spec}(\mathbb{Z}/p)$ and $\text{Spec}(\mathbb{Q})$ all consist of a single point.

Example 2: $\text{Spec}(\mathbb{Z})$ is the disjoint union of (p) for p prime and (0) . The quotient maps $\mathbb{Z} \rightarrow \mathbb{Z}/p$ corresponds to the inclusion $\text{Spec}(\mathbb{Z}/p) \rightarrow \text{Spec}(\mathbb{Z})$ where the unique prime ideal gets sent to the ideal (p) . The inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$ now gives the inclusion $\text{Spec}(\mathbb{Q}) \rightarrow \text{Spec}(\mathbb{Z})$ where the unique prime ideal gets sent to the ideal (0) .

Example 3: $\text{Spec}(\mathbb{C}[x, y])$ consists of 3 "kinds" of prime ideals: the ideal (0) , ideals of the form $(f(x, y))$ for some irreducible polynomial f , and the maximal ideals $(x - a, y - b)$. Geometrically, the maximal ideals correspond to points, the ideals (f) are "1-dimensional" points corresponding to the curves $f = 0$ in $\mathbb{C}^2$ and the ideal (0) corresponds to a "2-dimensional" point corresponding to the entire plane, as is discussed below.

Example 4/Exercise: For a prime ideal (f) as in Example 3, $\text{Spec}(\mathbb{C}[x, y]/(f))$ consists of two "kinds" of prime ideals: the "1-dimensional" ideal (0) and the "0-dimensional" points $(x - a, y - b)$ where $f(a, b) = 0$. The quotient map $\mathbb{C}[x, y] \rightarrow \mathbb{C}[x, y]/(f)$ gives the inclusion of $\text{Spec}(\mathbb{C}[x, y]/(f))$ into $\text{Spec}(\mathbb{C}[x, y])$ as the inclusion of an "abstract 1 dimensional curve" to a "curve embedded in the plane".

Now to clarify what is meant by dimension. A single point is always a 0 dimensional object, but points can "correspond" to higher dimensional subschemes by taking their closure in a certain topology. The intuition should be clear from example 3 however: for an ideal I , let $V(I)$ denote the set of all points at which *every* function in I vanishes. Then, $V((0))$ is the entire plane (2-dimensional), $V((f))$ is a 1-dimensional curve, and $V((x - a, y - b))$ is a single point. This is relevant in the arithmetic setting because the point (0) in $\text{Spec}(\mathbb{Z})$ is actually different from the other points - similar to example 4, it should be treated as a "generic point of a 1-dim. curve", that curve of course being $\text{Spec}(\mathbb{Z})$ itself.

To formalize this, we introduce the

Definition 4 (Zariski Topology). *The Zariski topology is a topology on $\text{Spec}(R)$ defined as follows: the closed sets correspond to $V(I) = \{\mathfrak{p} \in \text{Spec}(R) \mid I \subset \mathfrak{p}\}$, called the vanishing locus of I . The open sets are defined as $\text{Spec}(R) \setminus V(I)$ for some I . Note that for a prime ideal $\mathfrak{p} \in \text{Spec}(R)$, the topological closure is $V(\mathfrak{p})$.*

A few words are in order since the definition can seem a bit opaque. First, the condition $I \subset \mathfrak{p}$ is to be interpreted as " I is generated by functions which all vanish at $\mathfrak{p}$ ". Indeed, if $I = (f(x_1, \dots, x_n)) \subset \mathbb{C}[x_1, \dots, x_n]$ and $\mathfrak{p} = (x_1 - a_1, \dots, x_n - a_n)$ is a maximal ideal, then $I \subset \mathfrak{p}$ iff $f(a_1, \dots, a_n) = 0$. However, if we look at $\mathbb{R}[x, y, z]$, it is seen that $(x^2 + y^2 - 1) \subset (x^2 + y^2 - 1, z)$, the former corresponding to the unit sphere and the latter to the unit circle. So we see that the condition $I \subset \mathfrak{p}$ generalizes a function vanishing at a point to the statement $V(\mathfrak{p}) \subset V(I)$. The closure of a point is then seen to be all the points which "lie on that subscheme".

Secondly, notice that the open sets are very big. They are essentially the entire space minus the zero set of some polynomials. This is in contrast to our usual intuition with manifolds where open sets can be taken to be arbitrarily small. In particular, spectra are almost never Hausdorff as most pairs of open sets will have nonempty intersection. In case this definition seems like a strange way to define a topology, it should be noted that Poincaré initially formulated open sets of submanifolds of $\mathbb{R}^n$ as being sets which satisfied inequalities of the form $f > 0$ for some smooth function f on $\mathbb{R}^n$ (this can perhaps be seen as an early instance of algebraic geometry where the algebra of functions influences the geometry of a space). Another motivation is as follows: we want to base our geometry off of our algebra, so we should have a topology so that, for example, on $\text{Spec}(\mathbb{C}[x_1, \dots, x_n]) = \mathbb{C}^n$, we should have polynomials be continuous, and not really consider any other functions like e^x as continuous functions as they are not "algebraic" (note we want things to work over finite fields and things like $\mathbb{Z}$ over which such functions do not even make sense). It is then forced upon us that $V((f_1, \dots, f_k))$ is closed since it is the preimage of the origin, which should certainly be a closed set, under the map $(f_1, \dots, f_k) : \mathbb{C}^n \rightarrow \mathbb{C}^k$.

Exercise: Interpret this map $\mathbb{C}^n \rightarrow \mathbb{C}^k$ in the language of schemes, that is given functions $f_1, \dots, f_k \in \mathbb{C}[x_1, \dots, x_n]$, construct a map of schemes $\text{Spec}(\mathbb{C}[x_1, \dots, x_n]) \rightarrow \text{Spec}(\mathbb{C}[x_1, \dots, x_n])$ and verify this map is continuous in the Zariski topology. Hint: for continuity, use the equivalent formulation that the preimage of a closed set is closed.

Exercise: Show that for $\mathfrak{p}$ a prime ideal in R , the map $\text{Spec}(R/\mathfrak{p}) \rightarrow \text{Spec}(R)$ induced by the quotient $R \rightarrow R/\mathfrak{p}$ induces an inclusion of topological spaces. Actually, it is an isomorphism as topological spaces $\text{Spec}(R/\mathfrak{p})$ and $V(\mathfrak{p})$.

Hence, we can talk about the “dimensionality” of $\mathfrak{p}$ as the dimensionality of $\text{Spec}(R/\mathfrak{p})$. The good news is that there is a good algebraic description of dimension which sees an interplay between the geometry and the algebra.

Definition 5 (Krull Dimension). *The Krull dimension of a ring R is defined as any of the following equivalent statements:*

- The maximal integer n such that there is a strictly increasing chain of prime ideals $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n$ with $\mathfrak{p}_i \in \text{Spec}(R)$.
- The maximal integer n such that there is a strictly decreasing increasing sequence of vanishing sets $V(\mathfrak{p}_n) \subset \cdots \subset V(\mathfrak{p}_0)$ for $\mathfrak{p}_i \in \text{Spec}(R)$.
- (if R is a finitely generated algebra over a field K) The transcendence degree of $\text{Frac}(R)/K$, that is, the maximal number of K -algebraically independent functions in $\text{Frac}(R)$.

We say a scheme $\text{Spec}(R)$ is n -dimensional if R has Krull dimension n .

The first two definitions are restatements of the same thing while the third is a theorem (a citation can be found in the first chapter of [1]). The first is useful if knowledge of the ring is available. The second really provides the geometric intuition of the dimension: a sphere is two dimensional since you can find a point embedded in a circle embedded in the sphere. The third has the intuition that the number of algebraically independent functions is the number of “independent directions” of the scheme. This will be mentioned later, but algebraic extensions of rings correspond to maps of schemes with fibers a discrete set of points while transcendental extensions by one transcendental variable correspond to maps of schemes with fibers of a line. Note that we can interpret a ring as being Noetherian as saying it has finite Krull dimension.

Now we can define the “dimensionality of a point $\mathfrak{p} \in \text{Spec}(R)$ ” as the dimension of the quotient ring $R/\mathfrak{p}$. Note that this isn’t standard terminology, but is quite helpful for visualizations. We can also define the dimension as the dimension of the closure $V(\mathfrak{p})$ as follows: in the second definition, make $V(\mathfrak{p}_0) = V(\mathfrak{p})$ so that you only consider an increasing chain of subschemes up to the the subscheme that we care about. Equivalently, in the first definition, fix $\mathfrak{p}_0 = \mathfrak{p}$. Formally, what I am calling the “dimensionality of a point” is the same as the algebraic notion of the height of a prime ideal.

There are various ways to imagine the Zariski topology. One is to imagine a “total space” which would be a plane for $\text{Spec}(\mathbb{C}[x, y])$ or a curve for $\text{Spec}(\mathbb{Z})$ and draw the “ n -dimensional points” as being truly n -dimensional, that is, draw ideals like $(f) \in \text{Spec}(\mathbb{C}[x, y])$ as the curve $f = 0$. Another nice one is to have layers of points with each layer being the points of a certain dimension, with 0-dimensional points on the bottom and highest dimensional points on top. So for $\text{Spec}(\mathbb{C}[x, y])$, have the top layer be the single point (0) above a layer of points corresponding to (f) above a layer of points corresponding to maximal ideals (closed points). Then, draw lines between layers connecting $\mathfrak{p}$ and $\mathfrak{q}$ if $\mathfrak{p} \in V(\mathfrak{q})$ to get a poset of points denoting when one point is contained in the closure of another. This second picture is particularly useful for seeing what happens to generic points when you take open subsets or need to glue schemes together. Non-affine schemes are beyond these notes, but you can refer to just about any algebraic geometry textbook for examples.

3.3 Fibers

One very good way to think about a morphism of schemes is to look at the fibers of points in the image. Under this perspective, we can interpret a scheme map $\text{Spec}(S) \rightarrow \text{Spec}(R)$ as considering $\text{Spec}(S)$ a “ $\text{Spec}(R)$ parametrized family of schemes”. This will be useful for us since it will allow us to “probe” points of a scheme. In particular, we will later show how we can use this method to compute the prime ideals of a ring like $\mathbb{Z} \left[\frac{1+\sqrt{-3}}{2} \right]$

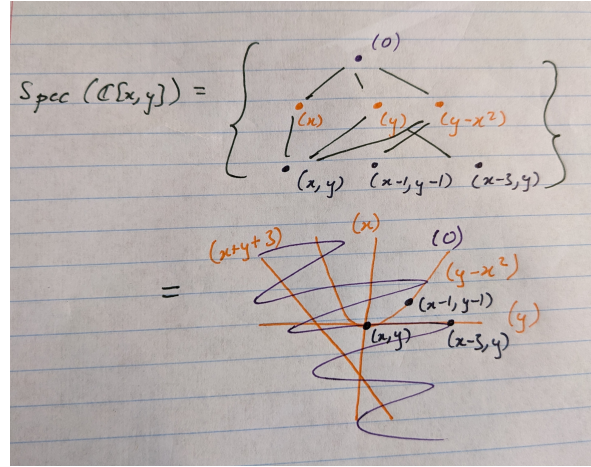


Figure 1: Two different ways to visualize schemes. The top is a poset displaying relationships with closures. The bottom shows every point as its closure.

3.3.1 Categorical Ideas

In topology, we have many instances of parametrized families of spaces over a base space. For example, we have (ramified) covering spaces, vector bundles, and more generally, fiber bundles. A very useful tool in studying these spaces is to pull them back to other spaces. That is, given a bundle $E \rightarrow X$, and given a map $f : Y \rightarrow X$, we can construct a fiber bundle f^*E where the fiber of f^*E over the point y is the fiber of E over the point $f(y)$. The picture can be summarized by the following commutative diagram:

$$\begin{array}{ccc} f^*E & \longrightarrow & E \\ \downarrow & & \downarrow \pi \\ Y & \xrightarrow{f} & X \end{array}$$

This is what we intuitively want from a pullback, but the construction is a bit more technical since we need to put a topology on f^*E to "connect points between fibers". Thus, it is not sufficient to just know the fibers of each point of Y , but we also need information between the fibers. The way that this is done is to set f^*E as the subset of points $p = (y, e) \in Y \times E$ such that $f(y) = \pi(e)$. The point is that $Y \times E$ has a natural topology and thus this induces a topology on f^*E . The interesting thing is that this construction is symmetric in (Y, f) and (E, π) , that is it doesn't really care which one is a vector bundle and which one is another map. Thus, we can also consider f^*E as a space with a map $\nu : f^*E \rightarrow E$ whose fibers $\nu^{-1}(e)$ have the same cardinality as $f^{-1}(\pi(e))$ (the preimage under the map $Y \rightarrow X$ of the point $x \in X$ which e lies over, thus there is a point mapping to e for each point mapping to x).

The fiberings in each direction give us a nice picture for f^*E , and we would like to generalize such a thing over arbitrary schemes. The important subtlety is that f^*E in this case was a topological space, but if Y, E, X are all affine schemes, we would like f^*E to also be an affine scheme, but it is not so obvious what ring this would correspond to. To find out, let's analyze the situation a bit closer. This diagram of a square actually comes up in more places that naturally arise geometrically.

Exercise: For U and V subsets of a set X and ι_U, ι_V the respective inclusions, show the following

diagram is commutative:

$$\begin{array}{ccc} U \cap V & \longrightarrow & U \\ \downarrow & & \downarrow \iota_U \\ V & \xrightarrow{\iota_V} & X \end{array}$$

Exercise: For X and Y arbitrary topological spaces, show the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & * \end{array}$$

where the bottom and right arrows are the unique maps that send every point to a single point $*$.

Exercise: For $\iota : x \rightarrow X$ being the inclusion of a point $x \in X$ and $f : Y \rightarrow X$ an arbitrary map, show the following diagram is commutative:

$$\begin{array}{ccc} f^{-1}(x) & \longrightarrow & Y \\ \downarrow & & \downarrow f \\ x & \xrightarrow{\iota} & X \end{array}$$

(The algebro-geometric analog of this example will actually be what is most useful for us).

Now these diagrams are a bit more special than just being commutative. Note that in each of the cases, if we take a proper subset of the top left corner, we still get a commutative diagram. However, what is special in all of these cases is that any other space that we can put in the top left corner *will* be a subset of what is currently there. That is, what we currently have in the top left corner will be the “maximal” (in categorical language, final) possible thing that we can have there.

Exercise: In each of the examples above, prove that if there is another set/topological space Z which fits into the top left corner, it has a unique map to what is currently in the top left corner.

This motivates the definition of the

Definition 6 (Categorical Product). *Given a category (e.g. topological spaces, sets, schemes) and a diagram*

$$\begin{array}{ccc} & Y_2 & \\ & \downarrow f_2 & \\ Y_1 & \xrightarrow{f_1} & X \end{array}$$

the categorical product of the diagram is an object Z , also denoted $Y_1 \times_X Y_2$, and a pair of maps g_i such that

$$\begin{array}{ccc} Z = Y_1 \times_X Y_2 & \xrightarrow{g_2} & Y_2 \\ \downarrow g_1 & & \downarrow f_2 \\ Y_1 & \xrightarrow{f_1} & X \end{array}$$

commutes and for any object W such that

$$\begin{array}{ccc} W & \xrightarrow{h_2} & Y_2 \\ \downarrow h_1 & & \downarrow f_2 \\ Y_1 & \xrightarrow{f_1} & X \end{array}$$

commutes, there is a unique map $W \rightarrow Z$. This is usually summarized by the following diagram:

$$\begin{array}{ccccc}
 & & W & & \\
 & \searrow & \downarrow h_1 & \nearrow h_2 & \\
 & & Z & \xrightarrow{g_2} & Y_2 \\
 & & \downarrow g_1 & & \downarrow f_2 \\
 & & Y_1 & \xrightarrow{f_1} & X
 \end{array}$$

Exercise: Verify each of the squares above is a categorical product.

3.3.2 Algebro-geometric Fiber Product

In our scenario, we would like to find a ring S' making the following diagram commute:

$$\begin{array}{ccc}
 \text{Spec}(S') & \longrightarrow & \text{Spec}(S_1) \\
 \downarrow & & \downarrow \\
 \text{Spec}(S_2) & \longrightarrow & \text{Spec}(R)
 \end{array}$$

when we know all the other spaces and the bottom and right maps. The two known maps of spectra are (by definition) induced by two ring homomorphisms $R \rightarrow S_i$ i.e. we have two R algebras S_1 and S_2 . Since maps between schemes and maps between rings are opposite one another, we may suspect that S' should satisfy the following commutative diagram:

$$\begin{array}{ccc}
 R & \longrightarrow & S_1 \\
 \downarrow & & \downarrow \\
 S_2 & \longrightarrow & S'
 \end{array}$$

with the finality of $\text{Spec}(S')$ translated into S' being initial for this diagram i.e. for any other ring S'' that makes this diagram commute, there is a unique map $S' \rightarrow S''$ (this is the categorical *coproduct*). The good news is that we already have a coproduct in the category of rings! It is the tensor product $S' = S_1 \otimes_R S_2$.

Definition 7 (Fibered Product of (Affine) Schemes). *Given affine schemes $Y_i = \text{Spec}(S_i)$, $i = 1, 2$ and $X = \text{Spec}(R)$ fitting into the following diagram:*

$$\begin{array}{ccc}
 & Y_2 & \\
 & \downarrow & \\
 Y_1 & \longrightarrow & X
 \end{array}$$

we define the fibered product $Y_1 \times_X Y_2$ to be the product of this diagram. It is given by $Y_1 \times_X Y_2 = \text{Spec}(S_1 \otimes_R S_2)$.

Note that implicitly there are a lot of ring homomorphisms in this definition (we need the initial data of maps $R \rightarrow S_i$ which make S_i into R algebras), but they are suppressed for ease of notation. They are however relevant in any calculation which must be done.

If $R = K$ is an algebraically closed field, say $\mathbb{C}$, and S_i are polynomial rings over K of dimension n_1, n_2 respectively, we see that their tensor product is a polynomial ring of dimension $n_1 + n_2$. More generally, if S_i are finitely generated algebras over $\mathbb{C}$, $\text{Spec}(S_i)$ would correspond to some complex manifolds and the fibered product over $\text{Spec}(\mathbb{C})$ would be the topological product of the manifolds

(in fact with the right complex structure). In general, the rough picture to have is that if we denote the maps $Y_i \rightarrow X$ as π_i , then the fibered product $Y_1 \times_X Y_2$ also has a map to X whose fiber over x is the topological products of $\pi_1^{-1}(x)$ and $\pi_2^{-1}(x)$. Compare this to the example of the pullback vector bundle above.

Let's consider some examples:

Example 1: Let's try and compute the prime ideals of $\mathbb{Z}[x]$. This may not seem so obvious at first, but we can understand these ideals by looking at the fibers of the map $\text{Spec}(\mathbb{Z}[x]) \rightarrow \text{Spec}(\mathbb{Z})$ induced by the inclusion $\mathbb{Z} \rightarrow \mathbb{Z}[x]$. For a maximal ideal $(p) \subset \mathbb{Z}$, the fiber is $\text{Spec}(\mathbb{Z}/p \otimes \mathbb{Z}[x]) = \text{Spec}(\mathbb{Z}/p[x])$ - the prime ideals of this ring are the generic (0) ideal and the maximal ideals $(f(x))$ for f an irreducible polynomial *mod* p . In $\mathbb{Z}[x]$, these correspond to (by taking pullbacks of ideals) (p) and $(p, f(x))$ respectively. Above the (0) ideal, we should look at the fiber of the inclusion $\text{Spec}(\mathbb{Q}) \rightarrow \text{Spec}(\mathbb{Z})$. Thus, we want to look at $\text{Spec}(\mathbb{Q} \otimes \mathbb{Z}[x]) = \text{Spec}(\mathbb{Q}[x])$ whose prime ideals are (0) and $(f(x))$ for an irreducible polynomial $f(x)$ *over* $\mathbb{Q}$. These correspond to the ideals in $\mathbb{Z}[x]$ (0) and ideals which consist of multiples of an irreducible polynomial over $\mathbb{Q}[x]$ that have coefficients in $\mathbb{Z}$. Note that the dimensionality of this space is 2 as we can find the increasing sequence of ideals $(0) \subset (p) \subset (p, f(x))$ or $(0) \subset (f(x)) \subset (p, f(x))$ (for suitable $f(x)$). We can see this as a family of one dimensional lines lying over a horizontal line $\text{Spec}(\mathbb{Z})$ similar to how we can think of a plane as a family of lines over a horizontal line. Consider the "zero set" of the polynomial $x^2 + 1$ on $\mathbb{Z}[x]$. Some maximal ideals at which this vanishes is $(2, 1 + x)$, $(3, x^2 + 1)$, $(5, x + 2)$, $(5, x - 2)$, so it can be seen as similar to the graph of $x = y^2$ in the plane.

Example 2: $\mathbb{Z} \rightarrow \mathbb{Z}[i]$ induces a map $\text{Spec}(\mathbb{Z}[i]) \rightarrow \text{Spec}(\mathbb{Z})$. The fiber of (0) corresponds to $\text{Spec}(\mathbb{Q} \otimes \mathbb{Z}[i]) = \text{Spec}(\mathbb{Q}(i))$, so there is one such point, namely (0) . Note that both are unique factorization rings. The fiber of the point (p) is $\text{Spec}(\mathbb{Z}/p \otimes \mathbb{Z}[i]) = \text{Spec}(\mathbb{Z}/p \otimes \mathbb{Z}[x]/(x^2 + 1)) = \text{Spec}(\mathbb{Z}/p[x]/(x^2 + 1))$. This splits into two ideals iff -1 is a square *mod* p . For example, $(1 + i)$ uniquely lies above (2) , $(2 + i)$ and $(2 - i)$ both lie above (5) and (3) is the unique prime ideal lying above (3) . We can interpret the primes lying above (p) as the primes in the prime factorization of p in $\mathbb{Z}[i]$. The above correspond to $2 = -i(1 + i)^2$, $5 = (2 + i)(2 - i)$, and 3 remaining prime in $\mathbb{Z}[i]$. Note that the quotient $\mathbb{Z}[x] \rightarrow \mathbb{Z}[x]/(x^2 + 1) = \mathbb{Z}[i]$ induces an inclusion of $\text{Spec}(\mathbb{Z}[i]) \rightarrow \text{Spec}(\mathbb{Z}[x])$ coinciding with the zero set of $x^2 + 1$ described in example 1.

Example 3: $\mathbb{Z} \rightarrow \mathbb{Z}[\sqrt[3]{2}]$. In order to factor primes, we need to study the equation $x^3 - 2$ modulo different primes. Above (2) , the equation simplifies to $x^3 = 0 \pmod{2}$ so we get $(2) = (2, \sqrt[3]{2})^3 = (\sqrt[3]{2})^3$. Above 3 , we have that $x^3 - 2 = x^3 + 1 = (x + 1)^3 \pmod{3}$ so that $(3) = (3, \sqrt[3]{2} + 1)^3$. Above 5 , $x^3 - 2 = (x - 3)(x^2 + 3x + 4)$, so $(5) = (5, \sqrt[3]{2} - 3)(5, (\sqrt[3]{2})^2 + 3\sqrt[3]{2} + 4)$. Above 31 , $x^3 - 2 = (x - 5)(x - 7)(x - 20)$ so $(31) = (31, \sqrt[3]{2} - 5)(31, \sqrt[3]{2} - 7)(31, \sqrt[3]{2} - 20)$ (to find this example, I took $7^3 - 2 = 341$ which had 31 as a factor which is $1 \pmod{3}$, and thus has a cube root of unity).

One thing to note is that the number of primes lying above a given prime seem to be bounded by the degree of the ring extension. In fact, we can say more. Let's assume for the time being the factorization of prime ideals in number rings. Then, say $\mathfrak{p}$ in the base ring A (in this case $\mathbb{Z}$) factors in the extension B as the product $\mathfrak{p}B = q_1^{e_1} \cdots q_n^{e_n}$. Then, we see that $\sum e_i$ in the above examples is also less than the degree of the extension. But we can say something even stronger: for a prime q_i appearing in the factorization of $\mathfrak{p}$, the composite map $A \rightarrow B \rightarrow B/q_i$ has kernel containing $\mathfrak{p}$, and so induces a map $A/\mathfrak{p} \rightarrow B/q_i$, which is a finite field extension. Let the degree of the extension be f_i . Then, we notice that in all the examples above, $\sum_{q_i|\mathfrak{p}} e_i f_i$ is the same as the degree of the extension (more precisely, of the field extension $\text{Frac}(B)/\text{Frac}(A)$).

Note that this is quite analogous to the fundamental theorem of algebra: there, we have that counted with multiplicity, the number of roots of $f(x)$ is $\sum e_i$ in an algebraically closed field like $\mathbb{C}$. In fact, we can instead look at $f(x) - c$ and see that this also has the same number of roots, counted with multiplicity. So, if we consider $f(x)$ as a map from $\mathbb{C} \rightarrow \mathbb{C}$, we see that every point has the same number of preimages, counted with multiplicities. This corresponds to the ring extension $\mathbb{C}[x] \subset \mathbb{C}[x][y]/(x - f(y)) = \mathbb{C}[y]$ (looking at the values of y where $f(y)$ is a certain value x). We don't see any f_i show up here, but in fact, if we used a non algebraically closed field and considered an equation like x^2 over $\mathbb{R}$, we would find the same thing holds if we consider the ring extension

$\mathbb{R}[x] \subset \mathbb{R}[x][y]/(x - f(y)) = \mathbb{R}[y]$ considered as schemes ($\text{Spec}(\mathbb{R}[x])$ still has morphisms from $\text{Spec}(\mathbb{C})$, but these correspond to $f_i = 2$). Thus, we see that this is really a generalization of the fundamental theorem of algebra.

3.4 Local Rings

Another very important concept in algebraic geometry is the concept of a local ring. We have seen that the order of divisibility of a prime in a principal ideal is analogous to how much the function vanishes at that point. This is a "local question" in the sense that it should not depend on the information of the function at points away from the one we are analyzing. In this section, we will discuss an algebraic construction called localization in which we can focus on a single point and remove the data of functions at points far away. Geometrically, the Spec of a local ring will have a unique maximal ideal, which is the point that we want to focus on.

Here are some places where we may want to discuss local questions in more classical situations:

- Power series converge in only a small disc. However, in this disc, it makes sense to discuss the order of vanishing of a function. For example, even though $\sum_{n=1}^{\infty} z^n$ only converges for $|z| < 1$ and not on all of $\mathbb{C}$, it makes sense to talk about an order of vanishing at 0 (in this case, it doesn't vanish).
- In similar vein, rational functions aren't defined everywhere, but it still makes sense to talk about orders of vanishing at points where they are defined. For example, $\frac{z^2(z-2)}{(z-1)(z-3)}$ vanishes at 0 to order 2.
- A different sort of question is as follows: when discussing how to obtain a topological map inducing a map of rings as its pullback, we discussed that given a map $F : X \rightarrow Y$ inducing a map $F^* : R(Y) \rightarrow R(X)$ on the function rings as pullbacks, we saw that if x_1 maps to y , $F^*(\mathfrak{m}_y) \subset \mathfrak{m}_{x_1}$. In particular, if we have multiple points, say x_1, x_2, x_3 which all map to y under F , $F^*(\mathfrak{m}_y) \subset \mathfrak{m}_{x_1} \mathfrak{m}_{x_2} \mathfrak{m}_{x_3}$, but we don't really care about the information of $\mathfrak{m}_{x_2}, \mathfrak{m}_{x_3}$, just that $F^*(\mathfrak{m}_y) \subset \mathfrak{m}_{x_1}$. It would be really convenient if this was actually an equality, and in fact, although it won't be discussed much here, it is precisely this sort of thing after restricting to the local rings which forms an essential part in the definition of a scheme.

There are two dual notions which give the intuition for local rings. First, the topological space associated to it (its Spec) only has a single point that you're focusing on (and other prime ideals "around it"). Algebraically, we are allowed to consider functions that aren't defined everywhere, but is still well defined at a single point. With this, we now define

Definition 8 (Local Ring). *A local ring is a ring with a unique maximal ideal.*

And to obtain such rings, we can use the process of

Definition 9 (Localization). *Given a ring R and a prime ideal $\mathfrak{p} \subset R$, we define the localization $R_{\mathfrak{p}}$ to be the ring R with formal inverses to the set $S = R \setminus \mathfrak{p}$. By adjoining a formal inverse to f , we mean we consider the ring $R[x]/(xf - 1)$. This is a local ring with unique maximal ideal $\mathfrak{p} R_{\mathfrak{p}} \subset R_{\mathfrak{p}}$. In the case that R is integral, $R_{\mathfrak{p}}$ is the subring of $\text{Frac}(R)$ where the denominators are exactly those elements of R not contained in $\mathfrak{p}$.*

The intuition for this definition is as follows: we can find an open subset of $\text{Spec}(R)$ by taking $\text{Spec}(R)[1/f]$ which is the complement of the primes at which f vanishes. We've seen, for example, that $\text{Spec}(\mathbb{Z}[1/6])$ has all the points of $\text{Spec}(\mathbb{Z})$ except for the ideals (2) and (3). Similarly, $\text{Spec}(\mathbb{C}[x] \left[\frac{1}{x^2-1} \right])$ would have all the prime ideals of $\mathbb{C}[x]$ except $(x+1)$ and $(x-1)$. The idea is that when we take Spec , we are creating a space on which some specified ring acts as the ring of functions. If we allow for rational functions, we must restrict to points on which the rational function is well defined (so that when we take R/I we aren't dividing by 0). Now, if we invert all functions which don't

vanish at a prime (point), what we are doing is eliminating all primes except the one we care about (and the primes contained in them) from the domain, leaving us with a unique maximal ideal.

A word of caution: if the ring we are working in has zero divisors, inverting by some elements may kill off other elements. For example, suppose we are working in $\mathbb{Z}/6$ where $2 \times 3 = 0$. If we invert by 3, then we have the equalities $2 = 2 \times 3 \times \frac{1}{3} = 0 \times \frac{1}{3} = 0$ (this is legal as we are adjoining an element $\frac{1}{3}$ to our ring). Thus, by inverting 3, we have inadvertently quotiented by 2. We will mostly be dealing with integral domains, but this example should highlight what happens with zero divisors and the reader is encouraged to try some examples of their own.

Example 1: $\mathbb{Z}_{(p)} = \{\frac{a}{b} \in \mathbb{Q} \mid p \nmid b\}$ is a 1 dimensional local ring with two prime ideals: the maximal ideal (p) and the generic ideal (0) .

Example 2: $\mathbb{Z}[\sqrt{-5}]_{(2, 1+\sqrt{-5})} = \{\frac{a}{b} \in \mathbb{Q}(\sqrt{-5}) \mid b \notin (2, 1+\sqrt{-5})\}$ is similarly a local ring with prime ideal (0) and unique maximal ideal $(2, 1+\sqrt{-5})$. Note that in the local ring, the ideal $(2, 1+\sqrt{-5})$ is actually principal! $2 = \frac{1-\sqrt{-5}}{3} \times (1+\sqrt{-5})$ so the ideal is actually equal to $(1+\sqrt{-5}) \mathbb{Z}[\sqrt{-5}]_{(2, 1+\sqrt{-5})}$.

Example 3: $\mathbb{C}[x, y]_{(x, y)}$ is a 2 dimensional local ring with maximal ideal (x, y) and prime ideals (0) (of height 0), and ideals like (x) , $(y - x^2)$, $(y^2 - x^3)$ of height one which are contained in (x, y) (i.e. corresponding to functions which vanish at $(0, 0)$)

Example 4: $\mathbb{C}[x, y]_{(x)}$ is a 1 dimensional local ring with ideals (0) and maximal ideal (x) .

To see what these rings look like in general, we state the following without proof (exercise)

Theorem 3.1. *The prime ideals of $R_{\mathfrak{p}}$ are of the form $\mathfrak{q} R_{\mathfrak{p}}$ for $\mathfrak{q} \subset \mathfrak{p}$. Any ideal I which is not contained in $\mathfrak{p}$ generates the whole local ring i.e. $IR_{\mathfrak{p}} = R_{\mathfrak{p}}$.*

The second statement essentially follows because I has some function which becomes inverted in the localization. The harder part to show is that this is a bijection. There is a nice picture of the localization of a ring R at a prime $\mathfrak{p}$ in the poset picture of $\text{Spec}(R)$: the prime ideals which lie in $\text{Spec}(R_{\mathfrak{p}})$ are precisely those which lie above the prime $\mathfrak{p}$. More precisely, the natural map $R \rightarrow R_{\mathfrak{p}}$ includes the points of $\text{Spec}(R_{\mathfrak{p}})$ to the primes lying above $\mathfrak{p}$. Compare this to $\text{Spec}(R/\mathfrak{p})$: the map $R \rightarrow R/\mathfrak{p}$ induces an inclusion of the points of $\text{Spec}(R/\mathfrak{p})$ to the primes of $\text{Spec}(R)$ lying below $\mathfrak{p}$ (i.e. the ideals of $R/\mathfrak{p}$ is in 1-1 correspondence with the ideals of R containing $\mathfrak{p}$). Thus, we see that geometrically, localizing is "dual" to quotienting. Oftentimes, you can analyze a space by looking "within a subscheme" by taking quotients and "around a subscheme" by taking the localization.

There are some very nice things that we can prove using local rings. Above, we saw that it appeared that for a "number ring" $\mathcal{O}_K$ of the field extension $K/\mathbb{Q}$, we have the identity $[K : \mathbb{Q}] = \sum_{\mathfrak{q}_i | \mathfrak{p}} e_i f_i$ where $\mathfrak{p} \subset \mathbb{Z}$ factors in $\mathcal{O}_K$ as $\mathfrak{p} = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_n^{e_n}$. It turns out we can prove this using local rings. The left hand side has to do with the field $\mathbb{Q}$ while the right hand side has to do with the prime $\mathfrak{p}$ i.e. the field $\mathbb{Z}/\mathfrak{p}$. A priori, these two fields seems disconnected e.g. their characteristics differ so they have very different behavior. But actually, it turns out that the local ring $\mathbb{Z}_{\mathfrak{p}}$ is able to "communicate" between $\mathbb{Q}$ and $\mathbb{Z}/\mathfrak{p}$.

For this argument, let's consider the inclusion $\text{Spec}(\mathbb{Z}_{\mathfrak{p}}) \rightarrow \text{Spec}(\mathbb{Z})$ and the fiber of $\mathcal{O}_K \rightarrow \mathbb{Z}$ of this inclusion. This will be the spectrum of the ring $S = \mathbb{Z}_{\mathfrak{p}} \otimes \mathcal{O}_K$ and will consist of the prime ideals of $\mathcal{O}_K$ lying above $\mathfrak{p}$ along with the ideal (0) . We can now look at the fibers of $\text{Spec}(\mathbb{Z}/\mathfrak{p}) \rightarrow \text{Spec}(\mathbb{Z}_{\mathfrak{p}})$ and $\text{Spec}(\mathbb{Q}) \rightarrow \text{Spec}(\mathbb{Z}_{\mathfrak{p}})$. In the first case, we get the spec of

$$\mathbb{Z}/\mathfrak{p} \otimes_{\mathbb{Z}_{\mathfrak{p}}} S = \mathbb{Z}/\mathfrak{p} \otimes_{\mathbb{Z}_{\mathfrak{p}}} (\mathbb{Z}_{\mathfrak{p}} \otimes \mathcal{O}_K) = (\mathbb{Z}/\mathfrak{p} \otimes_{\mathbb{Z}_{\mathfrak{p}}} \mathbb{Z}_{\mathfrak{p}}) \otimes_{\mathbb{Z}} \mathcal{O}_K = \mathbb{Z}/\mathfrak{p} \otimes \mathcal{O}_K = \mathcal{O}_K/\mathfrak{p} = \prod \mathcal{O}_K/\mathfrak{q}_i^{e_i}.$$

(The "associativity" of the tensor product of modules over different rings is a nontrivial lemma from commutative algebra that we will not prove). We can consider this ring as a $\mathbb{Z}/\mathfrak{p}$ module. It can be shown inductively (the argument will be given later) that $\dim_{\mathbb{Z}/\mathfrak{p}}(\mathcal{O}_K/\mathfrak{q}_i^{e_i}) = e_i f_i$. Thus $\prod \mathcal{O}_K/\mathfrak{q}_i^{e_i}$ has dimension $\sum e_i f_i$. Now, we have a very nice theorem about local rings, which we cite without proof:

Theorem 3.2 (Nakayama's Lemma). *For a local ring R with (unique) maximal ideal $\mathfrak{m}$ and a module M over R , if $m_1, \dots, m_n$ are elements of M such that the images of these elements form a basis of $M/\mathfrak{m}M$ as*

an $R/\mathfrak{m}$ vector field, then $m_1, \dots, m_n$ generate M as an R module. Alternatively said, if $\bar{m}_1, \dots, \bar{m}_n$ form a basis of $M/\mathfrak{m}M$ as an $R/\mathfrak{m}$ vector space, there exists elements m_i which reduce to $\bar{m}_i$ in $M/\mathfrak{m}M$ such that m_i generate M as an R module (in fact, any lifts will do).

Nakayama's lemma is often stated in more generality, but this version is a more intuitive statement, and (as far as I have seen), more useful. The take away is that general modules are very difficult to understand in general, but linear algebra over fields is easy. Nakayama's lemma allows us to partially understand a potentially difficult module over a (local!) ring R by understanding a vector space over the field $R/\mathfrak{m}$.

The way that we will use it in our case is that we can consider S as a module over the (local) ring $\mathbb{Z}_p$. We can treat $\mathbb{Z}/p \otimes_{\mathbb{Z}_p} S = \prod \mathcal{O}_K / \mathfrak{q}_i^{e_i}$ as S/pS when S is considered as a module. We demonstrated that this $\mathbb{Z}/p$ vector space has dimension $\sum e_i f_i$, and thus by Nakayama's lemma, there exist $\sum e_i f_i$ generators of S as a $\mathbb{Z}_p$ module. Now, we can look at the fiber over $\mathbb{Q}$:

$$\mathbb{Q} \otimes_{\mathbb{Z}_p} S = \mathbb{Q} \otimes_{\mathbb{Z}_p} (\mathbb{Z}_p \otimes \mathcal{O}_K) = (\mathbb{Q} \otimes_{\mathbb{Z}_p} \mathbb{Z}_p) \otimes_{\mathbb{Z}} \mathcal{O}_K = \mathbb{Q} \otimes \mathcal{O}_K = \text{Frac}(\mathcal{O}_K) = K$$

The second to last equality follows from being able to "rationalize" any element of $\text{Frac}(\mathcal{O}_K)$ to have denominator in $\mathbb{Q}$. On one hand, we showed $\mathbb{Q} \otimes_{\mathbb{Z}_p} S = K$, but on the other hand, we showed that S has $\sum e_i f_i$ generators, and thus we have K is a

$$\sum e_i f_i$$

dimensional vector space over $\mathbb{Q}$. There are some technical details that need to be checked such as none of the generators of S are torsion, but this is left to be worked out by the reader.

While this may have seemed unnecessary as the proof for extensions of $\mathbb{Z}$ could have been analyzed a bit more easily looking at degrees of polynomials reduced mod p , this proof generalizes (with essentially no change) to the scenario of having an arbitrary extension L/K and considering the extension of rings $\mathcal{O}_L/\mathcal{O}_K$ where it is less obvious that this is true if $\mathcal{O}_K$ is not a PID. In fact, the same proof can be used word for word to also prove the fundamental theorem of algebra, and more generally, analyze maps between algebraic curves (like from an elliptic curve over $\mathbb{C}$ to the Riemann sphere) to show that every point has the same number of preimages, counted with multiplicity.

3.5 An Interpretation of Modules

This is an optional subsection discussing a nice way to interpret modules over a ring. Similar to the previous proof, we can actually get a nice way to view arbitrary modules over an arbitrary commutative ring. Consider a ring R and an R module M . For every maximal ideal $\mathfrak{m} \subset R$, we can consider the module $M/\mathfrak{m}M$, and this naturally forms a vector space over the field $R/\mathfrak{m}$ (the action factors through $R/\mathfrak{m}$ since the action of $\mathfrak{m}$ is trivial). This is called the "fiber" of the module above $\mathfrak{m}$. Now, for a non-maximal prime ideal $\mathfrak{p}$, we can still consider the "fiber" $M/\mathfrak{p}M$ which forms a $R/\mathfrak{p}$ module. Now, since $\mathfrak{p}$ is prime, $R/\mathfrak{p}$ is in general only an integral domain, and $M/\mathfrak{p}M$ is still not as nice as a vector space. However, we can actually associate a field to $\mathfrak{p}$ as follows: the local ring $R_{\mathfrak{p}}$ is a ring with a unique maximal ideal, namely $\mathfrak{p}$. And thus, we can construct a field $R_{\mathfrak{p}}/\mathfrak{p}$. The natural map $R \rightarrow R_{\mathfrak{p}}/\mathfrak{p}$ gives an inclusion $\text{Spec}(R_{\mathfrak{p}}/\mathfrak{p}) \rightarrow \text{Spec}(R)$ which sends the unique point of the field to the prime $\mathfrak{p}$. You should also be able to get a field corresponding to $\mathfrak{p}$ by first quotienting by $\mathfrak{p}$ and then localizing. It's left as an exercise to see if these give the same field. If we consider $M \otimes_R R_{\mathfrak{p}}/\mathfrak{p}$, we get a vector space over this field! (Note that if $\mathfrak{m}$ is maximal, this is the same description as before).

Perhaps when you first learned modules, you learned it as "a vector space, but over a ring rather than a field". But this innocent looking statement hides a lot of the complications of modules. But now we see that we can consider our module M as being a family of vector spaces parametrized by different points (fields) of $\text{Spec}(R)$. Thus, it is something like a nonlinear family of vector spaces (similar to a vector bundle). Also similar to vector bundles, it is not enough to know the vector space above every point, you also need to know how these glue together i.e. you need to look at the family of vector

spaces over open sets (rings of the form $R[1/f]$). You can get a bit more information than the fiber (tensoring with the field) by looking at the stalk over a point (tensoring the module with the local ring $R_{\mathfrak{p}}$) - as was mentioned with Nakayama's lemma, we can get some information about this from the fiber.

4 Geometric Structure of Dedekind Domains

In this section, we will use the language of schemes that we have developed to elucidate the geometric structure of Dedekind domains and explain why it is the right context to have unique factorization of prime ideals. As a reminder, a Dedekind domain (what a "number ring" should be) is a ring A that satisfies the following axioms:

- A is Noetherian
- Every nonzero prime ideal of A is maximal
- A is integrally closed

In the language of schemes, the second condition that every nonzero ideal is maximal implies that $\text{Spec}(A)$ is a one dimensional scheme. Thus, it should be thought of as a curve. Now, any nonzero function will vanish only at maximal ideals, and hence a discrete set of points. The Noetherian condition implies that there is only a finite set of such points. As intuition, suppose a polynomial $f(x)$ vanished at infinitely many points $\alpha_1, \alpha_2, \dots$. Then, we can form the increasing chain of ideals

$$(f(x)) \subset \left(\frac{f(x)}{(x - \alpha_1)} \right) \subset \left(\frac{f(x)}{(x - \alpha_1)(x - \alpha_2)} \right) \subset \dots$$

A formal proof utilizes a technique called "Noetherian induction" and can be found in any standard algebraic geometry text.

Now the goal will be the following: for every prime (maximal) ideal $\mathfrak{p}_i$ at which $a \in A$ vanishes (i.e. $a \in \mathfrak{p}_i$), we want to define an "order of vanishing" $e_i \in \mathbb{N}$ analogous to how the order of vanishing n of $f(x)$ at α implies $(x - \alpha)^n \mid f(x)$. Then we consider the product ideal $I = \prod \mathfrak{p}_i^{e_i}$ (since a vanishes at finitely many primes, and we'll show e_i is finite, this product is well defined). We will then show $(a) = I$, which reduces to a local computation (both ideals vanish to the same order at each prime and this turns out to imply they're equal). In polynomial land, this says that the points and orders of vanishing of a polynomial in $\mathbb{C}[x]$ determines the polynomial up to a constant factor (and thus uniquely determines the ideal). This proves the existence of a prime factorization, and uniqueness follows from standard arguments.

The remainder of this section will be devoted to explaining the role of integral closure. It turns out that it only makes sense to define an order of vanishing (with the properties that we want) iff the ring is integrally closed. What we actually care about is that locally, every maximal ideal should correspond to a "nonsingular point". This is difficult to check at every maximal ideal, however, so we want a more "global" condition which locally agrees with this. Integral closure turns out to be the right thing to look at.

4.1 Singular Points

It turns out that although $\mathbb{Z}[i]$ and $\mathbb{Z}[\sqrt{-5}]$ have unique factorization of ideals, $\mathbb{Z}[\sqrt{-3}]$ doesn't! It turns out the correct ring (subring of $\mathbb{Q}(\sqrt{-3})$) to look at is $\mathbb{Z}\left[\frac{-1+\sqrt{-3}}{2}\right]$. The algebraic explanation is that $\mathbb{Z}[\sqrt{-3}]$ isn't integrally closed, but through some analysis of the quadratic formula, it can be shown that $\mathbb{Z}\left[\frac{-1+\sqrt{-3}}{2}\right]$ is. However, this does not feel satisfying since it doesn't explicitly display the failure

of UPFI in $\mathbb{Z}[\sqrt{-3}]$. It turns out the problem is at the ideal (2) (we will discuss later how I knew to look here).

You can check that the only prime ideal containing (2) is the ideal $\mathfrak{p} = (2, 1 + \sqrt{-3})$ (take fibers!). We see that $(2) \subsetneq \mathfrak{p}$ so we expect that $\mathfrak{p}^2 \mid (2)$ i.e. $2 \in \mathfrak{p}^2$ (e.g. $(x^2) \subsetneq (x)$). However, we find that $\mathfrak{p}^2 = (4, 2 + 2\sqrt{-3})$. Hence, we see that $\mathfrak{p}^2 \subsetneq (2) \subsetneq \mathfrak{p}$. This is an issue because we said that $\mathfrak{p}$ was the only prime dividing (2), so we should have that $2 = \mathfrak{p}^n$ for some n , but this n seems to lie between 1 and 2.

The problem can be geometrically interpreted as saying that the ideal $(2, 1 + \sqrt{-3})$ is a "singular point" of $\text{Spec}(\mathbb{Z}[\sqrt{-3}])$, just as the origin is a singular point of the curve $y^2 = x^3 - x^2$. A singular point of a curve in general is a point where a curve "intersects itself". This can be detected algebraically as follows: if the curve is of the form $f(x, y) = 0$, a singular point is a point of the curve where all partial derivatives vanish i.e. a point (x_0, y_0) where $f(x_0, y_0) = f_x(x_0, y_0) = f_y(x_0, y_0) = 0$. Some other examples are the point $(1, 1)$ on the curve $(x - 1)^2 - (y - 1)^2 = 0$ and the origin on the curve $y^2 = x^3$ (note that this does not appear to be a point of self intersection when looking over $\mathbb{R}$, but over $\mathbb{C}$, a neighborhood of the origin looks like the cone over the trefoil knot rather than a cone over an unknot i.e. a disk like a nonsingular point would look). We examined the curve $xy = 0$ in the introduction and noted that it's a bit subtle to analyze functions that vanish at the origin since they can either have a factor of x or y i.e. the function vanishing there does not give us information of a unique factor at that point. You may argue that the same is true at $(1, 0)$ with the functions $(x - 1)$ and y . However, since $xy = 0$, in fact we have that $y = -y(x - 1)$, so we can say that any function vanishing at $(1, 0)$ is a multiple of $(x - 1)$. However at $(0, 0)$, neither x nor y is a multiple of the other.

Now why should this be the algebraic condition for a curve having a self intersection? The idea is that if at least one of the partial derivatives is nonvanishing, we can locally parametrize the curve by the other variable: on the unit circle $x^2 + y^2 - 1 = 0$, the partial derivative w.r.t. x is nonvanishing at the point $(1, 0)$ and we can choose the y coordinate as a local coordinate, while at the point $(0, 1)$, the derivative w.r.t. y is nonvanishing, so we can locally parametrize with the x coordinate near this point. This is analytically the implicit function theorem, and we will later see an algebraic incarnation of this idea. Now, if the curve intersects itself, we cannot find a local parameter at the point, because it does not locally look like $\mathbb{R}$ (or $\mathbb{C}$ if working complexly). Thus, we must have that the derivative in every direction vanishes at a singular point.

Why are you intersecting yourself why are you intersecting yourself stop intersecting yourself stop intersecting yourself - Arie Ogranovich

This suggests that we should analyze derivatives in order to understand better how a function vanishing at a point determines a factor of the function. Another motivation for analyzing derivatives is that we use them to detect a double root. We can detect if $(x - a) \mid f(x)$ by checking if $f(a) = 0$, and if we know this, we can check if $(x - a)^2 \mid f(x)$ if in addition $f'(a) = 0$. In general, to check if $(x - a)^n \mid f(x)$, we can check if the first n derivatives of f vanish at a . We want to be able to say something similar in the case of number rings. Since our functions in this case are elements of our ring, we need to be able to take derivatives of numbers, but how could this possibly make sense?

We need to come up with an algebraic way to consider derivatives. One way to interpret the derivative is the best first order approximation to a function i.e. for x near a , $f(x) \approx f(a) + f'(a)(x - a)$ with an error of order $(x - a)^2$. One way to interpret this algebraically is to force $(x - a)^2 = 0$ so that we can replace the approximation with an equality. In other words, we should work in the ring $\mathbb{C}[x]/(x - a)^2$. There is one prime ideal in this ring, $(x - a)$, with associated field $\mathbb{C}[x]/(x - a) = \mathbb{C}$ and we said the image of $f(x)$ in this field is $f(a)$. Similarly, we can see (by using an eventually terminating Taylor series on polynomials) that the image of $f(x)$ in $\mathbb{C}[x]/(x - a)^2$ is $f(a) + f'(a)(x - a)$, and it can be checked that $f(a)$ and $f'(a)$ are the unique numbers for which we can write a representative of $f(x)$ in this form. Thus, we see that while $\mathbb{C}[x]/(x - a)$ keeps track of the value of a function at $x = a$, $\mathbb{C}[x]/(x - a)^2$ keeps track of the value of a function *and its derivative* at $x = a$.

Now, in order to define multiplicities, we want to ask whether a function which vanishes at a point described by a maximal ideal $\mathfrak{m} = (x - a)$ also has derivative zero. In other words, does a function $f(x) \in \mathfrak{m}$ map to 0 in $\mathbb{C}[x]/\mathfrak{m}^2$, or alternatively in $\mathfrak{m}/\mathfrak{m}^2$ (as $f \in \mathfrak{m}$). In $\mathbb{C}[x]$, $\mathfrak{m}/\mathfrak{m}^2$ is a 1 dimensional

$\mathbb{C}$ -vector space spanned by $(x - a)$, and the coefficient is exactly the derivative of a function in $\mathfrak{m}$. To test for higher multiplicities, we can test if $f(x) \in \mathfrak{m}^n$ (vanishes to degree n) also lies in $\mathfrak{m}^{n+1}$ (vanishes to 1 higher degree; the next derivative also vanishes), so we study $\mathfrak{m}^n / \mathfrak{m}^{n+1}$, which is a 1 dimensional vector space spanned by $(x - a)^n$ with the coefficient being the n th derivative. Note that taking quotients of powers of a maximal ideal make sense in any ring.

Let's then analyze $\mathfrak{m} / \mathfrak{m}^2$ in some generality: let's say we have a ring R with a maximal ideal $\mathfrak{m}$. We have that $\mathfrak{m}$ is naturally an R module and $\mathfrak{m} / \mathfrak{m}^2$ is then an $R / \mathfrak{m}$ vector space. This should correspond to the vector space of derivatives at the point $\mathfrak{m}$ in $\text{Spec}(R)$, but it also makes sense in an arithmetic context! This is known as the Zariski cotangent space at $\mathfrak{m}$ of $\text{Spec}(R)$. The tangent vector space is the space of tangent vectors, while the cotangent space is the space of (linear) functions on the tangent vectors. Tangent vectors in differential geometry can be interpreted as a small curve at a point, while cotangent vectors can be interpreted as first order approximations to functions restricted to such a curve i.e. the first order derivative of a function. The cotangent space is more amenable to algebraic geometry because a) we have access to functions and not infinitesimal curves and b) functions can naturally be added and scaled while the vector space structure of the tangent space in differential geometry relies on the existence of local trivializations and there is not a canonical coordinate-independent way to add them.

In case you're curious, you can define tangent vectors as follows: if a point $\mathfrak{p}$ of $\text{Spec}(R)$ has field $k = R_{\mathfrak{p}} / \mathfrak{p}$, we have an injection of $\text{Spec}(k) \rightarrow \text{Spec}(R)$ by the map $R \rightarrow k$ whose image is the point $\mathfrak{p}$. Now, a tangent vector is a map $\text{Spec}(k[x]/x^2) \rightarrow \text{Spec}(R)$ such that the composite $\text{Spec}(k) \rightarrow \text{Spec}(k[x]/x^2) \rightarrow \text{Spec}(R)$ is the map just specified. You can verify that such a map is equivalent to the data of a k -linear map $\mathfrak{p} / \mathfrak{p}^2 \rightarrow k$, where right hand side is the one dimensional vector space spanned by x . This element is the cotangent vector which pulls back to x , in differential geometry, a map from a curve to a manifold defines a pullback of 1 forms which is the linear map. The idea is that $\text{Spec}(k[x]/x^2)$ is an "infinitesimal neighborhood" of $\text{Spec}(k)$ - in differential geometry, we need to choose a small interval around the point, but there is an annoying equivalence condition to impose since it doesn't matter how small the interval is. Nicely, in algebraic geometry, we can bypass this problem since functions allow us to treat a single point as something like an infinitesimal neighborhood of another point. The vector space structure on tangent vectors comes from the linearity of the linear maps used to define the map of schemes. This already feels a bit more awkward to work with than the cotangent space because firstly, our vector space is a set of map of schemes and not really a vector space "living over" our point, and the linearity is more difficult to get to than that for the cotangent space which is found by literally adding functions which represent the cotangent vectors.

Example: Say we're looking at the curve $f(x, y) = 0$ and for simplicity suppose $(0, 0)$ lies on the curve and is a nonsingular point (for any other nonsingular point (a, b) , we can instead consider the function $f(x - a, y - b)$ to reduce to this case). We will analyze the cotangent space at $\text{Spec}(\mathbb{C}[x, y]/f) \subset \text{Spec}(\mathbb{C}[x, y])$, which is the curve cut out by $f = 0$, at the ideal $(x - 0, y - 0)$. Call $R = \mathbb{C}[x, y]/f$ and $\mathfrak{m} = (x, y)$. Suppose that $f_x(0, 0) = c \neq 0$ and $f_y(0, 0) = d$ which may or may not be 0. Then, as $f(0, 0) = 0$, the image of $f(x, y)$ in $\mathbb{C}[x, y] / \mathfrak{m}^2$ (i.e. the first order approximation) is $f(x, y) = cx + dy$. To look at the cotangent space, we should look at $R / \mathfrak{m}^2 = \mathbb{C}[x, y] / (x, y, f)$, so in this ring, we also have $cx + dy = 0$, and since c is nonzero, we can write $x = -dy/c$. Thus, we see $\mathfrak{m} / \mathfrak{m}^2$ considered as an $R / \mathfrak{m}$ vector space is principally generated by y ($\mathfrak{m}$ is generated by x and y as a $\mathbb{C}$ vector space and x is a multiple of y). Thus, we see that the cotangent space to a curve at a nonsingular point is one dimensional, as expected since the tangent space is one dimensional. Note that we could've worked over any finite field, even those of finite characteristic. Also note the similarity of this with the implicit function theorem, more on this later.

Exercise: For a curve $f(x, y) = 0$ which is singular at the origin e.g. $xy = 0$ or $y^2 - x^3 - x^2 = 0$, show that the cotangent space at the ideal (x, y) of $\text{Spec}(\mathbb{C}[x, y]/f)$ is a two dimensional $\mathbb{C}$ vector space.

Exercise: Show that the cotangent space of $\text{Spec}(\mathbb{R}[x, y])$ at a maximal ideal $\mathfrak{m}$ is a two dimensional K vector space where $K = \mathbb{R}[x, y] / \mathfrak{m}$ is either $\mathbb{R}$ or $\mathbb{C}$. Do an example for each.

Exercise: Show that the cotangent vector space of $\text{Spec}(\mathbb{Z}[i])$ at a maximal ideal $\mathfrak{m}$ is a one dimensional $\mathbb{Z}[i] / \mathfrak{m}$ vector space. Do an example with $\mathfrak{m} = (p)$ and $\mathfrak{m} = (a + bi)$

Exercise: Show that the cotangent space of $\text{Spec}(\mathbb{Z}[\sqrt{-5}])$ at $\mathfrak{m} = (2, 1 + \sqrt{-5})$ is a one dimensional $\mathbb{Z}/2$ vector space (hint: factor 6).

Exercise: Show the cotangent vector space of $\text{Spec}(\mathbb{Z}[\sqrt{-3}])$ at $(2, 1 + \sqrt{-3})$ is a two dimensional $\mathbb{Z}/2$ vector space.

Exercise: Show that the cotangent vector space of $\mathbb{C}[z_1, \dots, z_n]$ at any maximal ideal is an n -dimensional $\mathbb{C}$ vector space.

Exercise: Take k functions $f_1, \dots, f_k \in \mathbb{C}[z_1, \dots, z_n]$ such that the Jacobian matrix has maximal rank (of k) at a point $(a_1, \dots, a_n)$. Show that the Zariski cotangent space of $\mathfrak{m} = (z_1 - a_1, \dots, z_n - a_n)$ in $\text{Spec}(\mathbb{C}[z_1, \dots, z_n]/(f_1, \dots, f_n))$ has $\mathbb{C}$ dimension $n - k$.

Exercise: Let $\mathfrak{m}$ be a maximal ideal of $\text{Spec}(\mathbb{Z}[x])$ lying over the ideal (p) of $\text{Spec}(\mathbb{Z})$. Show that $\mathfrak{m}/\mathfrak{m}^2$ is a 2 dimensional $\mathbb{Z}/p$ vector space.

Exercise: Pick any of the rings R above (or your favorite ring) and pick a non-maximal prime ideal $\mathfrak{p}$. Compute the dimension of the cotangent space $\mathfrak{p}R_{\mathfrak{p}}/\mathfrak{p}^2R_{\mathfrak{p}}$. (Why should this be the correct definition? What is the field that you should consider this a vector field over?) What is the height of the ideal $\mathfrak{p}$ that you chose?

We see that in polynomial cases, when we have a nonsingular point, the dimension of the Zariski cotangent space is equal to the dimension of $\text{Spec}(R)$. This is also true in most arithmetical settings, but fails for the case of $(2, 1 + \sqrt{-3})$, where we saw an issue with unique prime factorization. Thus, we get the sense that the existence of a well-defined integer multiplicity has to do with a point being "nonsingular" in some sense relating to the Zariski cotangent space. We thus make the following definition:

Definition 10 (Regular Point). *A maximal ideal $\mathfrak{m}$ of $\text{Spec}(R)$ is called a regular or nonsingular point if the Zariski cotangent space $\mathfrak{m}/\mathfrak{m}^2$ as an $R/\mathfrak{m}$ vector space has the same dimension as the dimension of $\text{Spec}(R)$. A point which is not regular is called singular.*

The idea is that the dimension of the tangent space (and hence the cotangent space) should measure the "number of directions" accessible/tangent at that point. For a singular point of a curve, we can move in at least 2 directions, which is more than the 1 dimension of a curve. It should be noted that the dimension of the cotangent space at a point is at least the dimension of $\text{Spec}(R)$ (the idea will be explained later), so that a singular point has the dimension of the (co)tangent space strictly greater than the dimension of $\text{Spec}(R)$.

Now, we explain the relevance of the cotangent space on defining multiplicities of vanishing of functions (elements of our ring). The idea, alluded to earlier, is that computing the order of vanishing is really a local question, and so if we want to analyze the order of vanishing of f at the point $\mathfrak{m}$, we can restrict our attention to the local ring $R_{\mathfrak{m}}$. We state the following without proof (we ignore the integrally closed condition until the next subsection):

Theorem 4.1. *The following are equivalent for a local dimension one ring $\mathfrak{R}$ with maximal ideal $\mathfrak{m}$:*

- $\mathfrak{m}/\mathfrak{m}^2$ is a one dimensional $\mathfrak{R}/\mathfrak{m}$ vector space.
- $\mathfrak{m}$ is a principal ideal.
- $\mathfrak{R}$ is a PID (in particular, every ideal is principal).
- $\mathfrak{R}$ is a UFD.
- If π is a generator for $\mathfrak{m}$ i.e. $\mathfrak{m} = (\pi)$, any ideal of $\mathfrak{R}$ is of the form $(\pi^n) = (\pi)^n = \mathfrak{m}^n$. Such a π is called a uniformizer.
- $\mathfrak{R}$ is integrally closed.

Such rings are called discrete valuation rings (DVRs).

The statement that every ideal is of the form $(\pi)^n$ is the one that we care about: suppose $\mathfrak{m}$ is a regular point of $\text{Spec}(R)$ where the element ("function", but also works arithmetically) $\alpha \in R$ vanishes. Then, we can analyze the ideal $\alpha R_{\mathfrak{m}}$ (in the local ring $\mathfrak{A} = R_{\mathfrak{m}}$) and by this theorem, it must be equal to $\mathfrak{m}^n$ for some n , and so we define the order of vanishing of α at $\mathfrak{m}$ to be n . The idea is that any uniformizer π (note any two differ by a unit) serves as a local parameter e.g. $(x - a)$ and since $(\alpha) = (\pi^n)$, we have $\alpha = \pi^n u$ for some unit u in the local ring $R_{\mathfrak{m}}$ i.e. some function that doesn't vanish at $\mathfrak{m}$. We can also analyze this in terms of looking at the image of α in a filtration $\mathfrak{m}^n / \mathfrak{m}^{n+1}$ of $\mathfrak{m}$. Note that each $\mathfrak{m}^n / \mathfrak{m}^{n+1}$ is a one dimensional $R/\mathfrak{m}$ vector space spanned by π^n (the coefficient being the " n th-derivative with respect to the local parameter π "). Then we have that there must be some minimal n where the derivative is nonzero, and this determines the multiplicity.

Now to explain some of the relationships of the theorem: the fact that $\mathfrak{A}$ as in the theorem is a PID and DVR follows, in a bit of a technical manner, from the maximal ideal $\mathfrak{m}$ being principal [2]. Now, the number of generators of the maximal ideal is quite directly related to the Zariski cotangent space! This is another application of Nakayama's lemma: $\mathfrak{m}$ can be considered as an $\mathfrak{A}$ module and then, by Nakayama, a (minimal) basis for $\mathfrak{m} / \mathfrak{m}^2$ as an $\mathfrak{A} / \mathfrak{m}$ vector space lifts to a set of generators for $\mathfrak{m}$ as an $\mathfrak{A}$ module. Now, if $\mathfrak{m}$ had fewer than $\dim(\mathfrak{m} / \mathfrak{m}^2)$ generators, then reducing mod $\mathfrak{m}$, $\mathfrak{m} / \mathfrak{m}^2$ would have a smaller basis, contrary to assumption. This argument does not rely on $\mathfrak{A}$ being one dimensional, and actually works in general. Thus, we see that if $\mathfrak{A}$ is a local n dimensional ring, the cotangent space must be at least n dimensional, as was previously claimed. This is an algebraic incarnation of the geometric principle that the tangent space should model the geometry locally, and thus dually, the cotangent vector space should model functions locally. Indeed, we see that elements of the cotangent space lift to functions locally defined.

4.2 Proof of Unique Prime Factorization

We now finish the proof of unique prime factorization using some more theorems whose proofs will be omitted. They are not extremely difficult proofs, but distract from the main content. First, we note that similar to how we can formally invert elements of a ring, we can also formally act on modules by inverted elements. In this spirit, we define $M_{\mathfrak{m}} = R_{\mathfrak{m}} \otimes M$ to be the localization of the module M at the point $\mathfrak{m}$ (note this minimally forces an action of elements invertible in $R_{\mathfrak{m}}$ on M , or at least something related to it). To give some indication as to what this does, it kills off any element of m which is torsion w.r.t. any element $a \in R \setminus \mathfrak{m}$ i.e. $am = 0$ since $m = am \times \frac{1}{a} = 0$. We can think of $M_{\mathfrak{m}}$ as those elements of M which aren't killed (i.e. non-torsion, or m such that the annihilator is contained in $\mathfrak{m}$) with coefficients in $R_{\mathfrak{m}}$.

Theorem 4.2. *The following all hold [2]:*

- If $M \rightarrow M' \rightarrow M''$ is an exact sequence of R modules, $M_{\mathfrak{m}} \rightarrow M'_{\mathfrak{m}} \rightarrow M''_{\mathfrak{m}}$ is an exact sequence of $R_{\mathfrak{m}}$ modules. This statement holds more generally with the more general notion of localization, but we won't use it here.
- If $\phi : M \rightarrow N$ is a map of R -modules such that the induced map $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$ is an isomorphism for all maximal ideals $\mathfrak{m}$ of R , we have that ϕ itself is an isomorphism.
- For an integral domain R , $R = \bigcap_{\mathfrak{m} \in \text{Specm}(R)} R_{\mathfrak{m}}$ i.e. R is the intersection of all the localizations at maximal ideals. Note that the intersection makes sense since, as R is integral, $R_{\mathfrak{m}} \subset \text{Frac}(R)$.

The first statement says that an exact sequence of modules (which can be thought of as functions with codomains changing from point to point) remains exact when we restrict from looking over all of $\text{Spec}(R)$ to just $\text{Spec}(R_{\mathfrak{m}})$. The second says that in order to check if a morphism of modules is an isomorphism, it suffices to check locally. The third can be interpreted as follows: $R_{\mathfrak{m}}$ was constructed by inverting some functions, and if these functions vanished somewhere, their reciprocal would have a "pole" at that point (this can be formalized at nonsingular points by considering "fractional ideals")

and extending the valuation to a $\mathbb{Z}$ valued function). Thus the local ring at $\mathfrak{m}$ is the functions “well defined at $\mathfrak{m}$ ” (not having poles). This statement says that functions well defined everywhere must have lied in our original ring to begin with i.e. inverting anything that wasn’t already inverted must introduce poles i.e. any non-unit must vanish somewhere i.e. any function that vanishes nowhere is already invertible.

Finally, we prove our main theorem. Consider some ring R that is a dimension 1 Noetherian ring (the first two conditions of a Dedekind domain) with the property that at every maximal ideal $\mathfrak{p}$, $R_{\mathfrak{p}}$ is a DVR (I will use $\mathfrak{p}$ instead of $\mathfrak{m}$ here since it corresponds to primes of a number ring). For an ideal $I \subset R$, looking locally at every prime $\mathfrak{p}_i$, we have $IR_{\mathfrak{p}_i} = \mathfrak{p}_i^{e_i} R_{\mathfrak{p}_i}$ (note this is only possible because $R_{\mathfrak{p}_i}$ is a DVR!). Consider then the product $J = \prod \mathfrak{p}_i^{e_i} \subset R$ (as remarked before, the 1-dimensionality + Noetherian property means this product is finite). Note that

$$JR_{\mathfrak{p}_j} = \left(\prod \mathfrak{p}_i^{e_i} \right) R_{\mathfrak{p}_j} = \prod \left(\mathfrak{p}_i^{e_i} R_{\mathfrak{p}_j} \right) = \mathfrak{p}_j^{e_j} R_{\mathfrak{p}_j}.$$

Now, as $I \subset IR_{\mathfrak{p}_i} \subset JR_{\mathfrak{p}_i}$ (as R modules) for all i , we have that

$$I \subset \left(R \cap \bigcap JR_{\mathfrak{p}_i} \right) = \bigcap (R \cap JR_{\mathfrak{p}_i}) = \bigcap (R \cap \mathfrak{p}_i^{e_i} R_{\mathfrak{p}_i}) = \bigcap \mathfrak{p}_i^{e_i} = \prod \mathfrak{p}_i^{e_i} = J$$

where the second to last inequality follows from the (nontrivial) fact that for any ideal $K \subset R$, $K = R \cap KR_{\mathfrak{p}}$ for any ideal $\mathfrak{p}$ and the last equality follows from the Chinese remainder theorem. The takeaway is just that $I \subset J$ and so we get an R module homomorphism which is the inclusion $\phi : I \rightarrow J$. We wish to show this is an equality. But by the above theorem, this can be checked at every maximal ideal, and at every maximal ideal, $IR_{\mathfrak{p}_i} = JR_{\mathfrak{p}_i} = \mathfrak{p}_i^{e_i} R_{\mathfrak{p}_i}$, so ϕ is locally an isomorphism at every maximal ideal, and is thus an isomorphism. And hence $I = \prod \mathfrak{p}_i^{e_i}$!

Note that this proof works in two notable cases: integral closures of $\mathbb{Z}$ in finite extensions of $\mathbb{Q}$ (as will be seen momentarily) and nonsingular curves over finite fields e.g. $\mathbb{Z}/p\mathbb{Z}[x, y]/(y^2 - x^3 - x)$.

4.2.1 The Role of Integral Closure

Now, we never actually used integral closure in our proof, so why is it relevant? Well, we used that the localization at *every* maximal ideal is a DVR, but this is difficult to check! After all, there’s infinitely many primes. We would like a way to check if localizing at every maximal ideal of R gives a DVR with something intrinsic to R i.e. not have to actually check every local ring. Note that even though a maximal ideal may be principal in the local ring, it may not be principal in R e.g. $(2, 1 + \sqrt{-5})$ in $\mathbb{Z}[\sqrt{-5}]$ (locally generated by $(1 + \sqrt{-5})$ as was seen earlier); in fact, this was the whole reason to look locally! Thus, “an ideal being principal” does not translate from a local to a global condition. However, we stated in the theorem that locally, a ring is a DVR iff it is integrally closed. From the fact that $R = \bigcap_{\mathfrak{m} \in \text{Specm}(R)} R_{\mathfrak{m}}$, it can be checked (exercise!) that R is integrally closed iff the localization at every maximal ideal is integrally closed.

Thus, although a point being nonsingular i.e. the local ring being a PID/DVR is what we *really* care about, this is not something that can be easily related from a local to a global property. However, the property of integral closure can be made global! And thus, we can check algebraically if every local ring is a DVR without every having to localize. Ironically, however, it is often very useful to just check for integral closure (nonsingularity) locally anyway. We have a fairly easy way to do this: just check the dimension of the cotangent space $\mathfrak{m}/\mathfrak{m}^2$. Integral closure can be explicitly checked for simple cases like quadratic extensions, but try checking whether a rank 5 extension of $\mathbb{Z}$ is integrally closed, doesn’t seem so easy now does it?

4.3 A Bit About Discriminants

The main difficulty in checking nonsingularity pointwise is to know which maximal ideals to look for (and thus not have to check every single point). For number rings, there is a useful construction called

the discriminant (the definition can be found in any introductory algebraic NT text such as [3], [4], or [2]) which essentially checks "relative nonsingularity over a base", and it turns out a prime in the base ring divides the discriminant iff the fiber contains a ramified prime or a singular prime. This eliminates all but finitely many maximal ideals to check!

Thus, we have an algorithmic way (in principle) to check for the integral closure of $\mathbb{Z}$ in any finite extension $K/\mathbb{Q}$:

- First, find a primitive generator for $K/\mathbb{Q}$ i.e. an element $\alpha \in K$ such that $K = \mathbb{Q}(\alpha)$ - in fact, "most" elements are primitive.
- If α is not integral i.e. the leading coefficient of the minimal polynomial $f(x) \in \mathbb{Z}[x]$ does not have leading coefficient 1, replace α with $n\alpha$ for sufficiently large (in terms of divisors) $n \in \mathbb{Z}$ so that the minimal polynomial of $n\alpha$ is monic (it is left as an exercise to show this). Note $n\alpha$ is still primitive.
- Now consider the ring $A = \mathbb{Z}[n\alpha]$.
- Compute the discriminant Δ of A over $\mathbb{Z}$.
- For primes p dividing Δ , compute the primes $\mathfrak{p}_i$ lying above p by analyzing $A \otimes \mathbb{Z}/p\mathbb{Z}$.
- Compute $\mathfrak{p}_i / \mathfrak{p}_i^2$ for each of these primes, and check if it forms a 1 dimensional $A / \mathfrak{p}_i$ vector space.
- If it isn't find some element in $\text{Frac}(A)$ which is integral over $\mathbb{Z}$ and adjoin it to A (an example is given below).
- Repeat steps 4 to 7 until every maximal ideal is a regular point.

The reason you want to consider integral extensions is so that every ideal over the base ($\text{Spec}(\mathbb{Z})$ in this case) has nonempty fiber e.g. $\mathbb{Z}[1/6]$ does not have primes above 2 and 3, and neither does $\mathbb{Z}[i/6]$, where $i/6$ is not integral. That integral extensions always have nonempty fibers is guaranteed by the "going up theorem". Note that if you chose a very large n in step 2, (e.g. took $\mathbb{Z}[2i]$), you would have to "divide through" in steps 4-7 (the idea is that more functions can "distinguish" more points and hence make things nonsingular, while conversely, not enough functions can't distinguish points and may identify two points, thus creating a singularity).

Example: Let's finally analyze $\mathbb{Z}[\sqrt{-3}]$ in some detail. The discriminant turns out to be

$$\left(\det \begin{pmatrix} 1 & \sqrt{-3} \\ 1 & -\sqrt{-3} \end{pmatrix} \right)^2 = -12$$

and thus the only possibilities for a singularity is a prime above 2 or a prime above 3. 3 factors as $(\sqrt{-3})^2$, and the ideal $\mathfrak{p} = (\sqrt{-3})$ is already principal, so it's not hard to convince yourself that $\mathfrak{p} / \mathfrak{p}^2$ will be one dimensional. However, $2 = (2, 1 + \sqrt{-3})^2$ and so we need to check if $\mathfrak{q} = (2, 1 + \sqrt{-3})$ is regular. We see that $\mathfrak{q} / \mathfrak{q}^2$ is 2 dimensional since neither generator can be written as a multiple of the other without dividing by something in the ideal (exercise; contrast with $(2, 1 + \sqrt{-5})$ we saw earlier, the condition of $p \pmod{4}$ when finding integral closures in $\mathbb{Q}(\sqrt{-p})$ should make sense from this point of view!). If we set $x = 1 + \sqrt{-3}$ and $y = 2$, we find that $x^2 = -2 + 2\sqrt{-3} = 2x - 4 = xy - y^2$. Notice that the equation $y^2 - xy + x^2 = 0$ is singular at 0, this can be used to explain the singularity here (this is what the "curve" looks like near this point, and you can't find a local generator with this). However, we see from this that $z = \frac{x}{y}$ satisfies $(\frac{x}{y})^2 - \frac{x}{y} + 1 = z^2 - z + 1 = 0$. Then $z \in \text{Frac}(\mathbb{Z}[\sqrt{-3}])$ and is integral over $\mathbb{Z}$. This trick of relating two generators of the ideal by an algebraic equation and dividing through by some power of one of the variables to get a monic equation in their ratio works quite generally to "desingularize" curves.

We recognize here that z is a sixth root of unity i.e. $\mathbb{Z}[\sqrt{-3}][z] = \mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$. An here, (2) factors as $(2) = \left(2, \frac{1+\sqrt{-3}}{2}\right) \left(2, \frac{1-\sqrt{-3}}{2}\right)$. Both of these are regular as to check if $\left(2, \frac{1+\sqrt{-3}}{2}\right)$ is regular, we

have $2 = \frac{1+\sqrt{-3}}{2} \left(\frac{1-\sqrt{-3}}{2} \right)$, and thus the ideal is locally generated by $\frac{1+\sqrt{-3}}{2}$. Or alternatively, if we computed the discriminant of this ring, we would find it was -3 , so we never even had to check 2! Notice that since we "didn't have enough functions", two points got identified into one point which became a singularity - the two "directions" of the points that were identified meant that the resulting combined point had a 2 dimensional tangent space.

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